

Towards Qualitative Measurement at Scale: A Prompt-Engineering Framework for Large-Scale Analysis of Deliberative Quality in Parliamentary Debates

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ABSTRACT

This paper investigates whether Large Language Models (LLMs), guided by prompt engineering, can automate the complex Discourse Quality Index (DQI) for measuring deliberative quality. Evaluating state-of-the-art models from OpenAI and Google by varying In-Context Learning (ICL) examples (0–50) shows LLMs achieve high fidelity, comparable to human annotators based on standard reliability metrics. Performance improves markedly with few examples, plateauing around 25–50 examples. While both models perform well, differences highlight the interplay between model selection and ICL strategy. Error analysis identifies specific DQI dimensions requiring further improvement, suggesting future work on advanced reasoning prompts. This study confirms LLM viability for scaling DQI measurement and provides practical guidance on optimizing ICL strategies. Additionally, it contributes a modular, adaptable AI engineering pipeline that researchers can leverage for their own prompting experiments across various measurement tasks.

Keywords: Discourse analysis, large language models, artificial intelligence, Congress

1 Introduction

The ability to measure complex linguistic, psychological, and social constructs from textual data is central to many research questions in the social sciences and humanities. In political science, for example, understanding the quality of deliberative communication in parliamentary debates is essential for evaluating the health of democratic institutions and the legitimacy of political decisions (Habermas, 1996; Bächtiger *et al.*, 2018; Steenbergen *et al.*, 2003). Similarly, tracking the evolution of social norms and cultural beliefs over time involves measuring latent dimensions from large historical corpora (Garg *et al.*, 2018; Grossmann and Varnum, 2015).

Traditionally, measuring such rich, high-dimensional constructs from text has relied on qualitative methods like expert human annotation. Approaches like the Discourse Quality Index (DQI, Steenbergen *et al.*, 2003) leverage the knowledge and interpretive abilities of trained coders to make judgments about multiple dimensions of textual data. While these methods can yield valid and reliable measurements, they are often prohibitively expensive and time-consuming to apply to large datasets. As a result, many researchers have turned to more scalable computational methods, such as dictionary-based analysis (Tausczik and Pennebaker, 2010), supervised machine learning (Rudkowsky *et al.*, 2018), and unsupervised topic modeling (Blei *et al.*, 2003). However, these automated methods typically measure relatively narrow constructs, and often struggle to capture the deeper meaning and context embedded in text.

Recent advances in Large Language Models (LLMs) offer a promising pathway to overcome this enduring trade-off between measurement richness and scalability (Brown *et al.*, 2020). These models, pre-trained on vast datasets, possess remarkable capabilities for understanding and generating human language. Through prompt engineering—the process of designing instructions to guide model behavior—researchers can leverage LLMs to perform complex analytical tasks (Liu *et al.*, 2021; Xiao *et al.*, 2023; Ornstein *et al.*, 2025). However, understanding how to reliably and efficiently elicit high-quality annotations comparable to human experts remains a research challenge (Bojic *et al.*, 2025), particularly for established, complex social science measures (Halterman and Keith, 2025).

This paper uses the DQI (Steenbergen *et al.*, 2003) as a case study to investigate the potential for generative AI models to automate expert annotation of political texts. The DQI is a widely respected framework in political science for assessing the deliberative quality of political communication (Habermas, 1996; Bächtiger *et al.*, 2018). Its multi-dimensional nature, reliance on contextual interpretation, and ordinal scoring scales make it notoriously labor-intensive to apply manually, limiting the extent to which it can be used to analyze

large corpora of political speech. Automating the DQI therefore represents a significant test for LLM capabilities in complex, theory-driven measurement.

While prior work suggests LLMs can perform qualitative coding tasks (Xiao *et al.*, 2023; Omizo, 2024), this paper makes the following contributions: First, I evaluate the impact of In-Context Learning (ICL) example quantity on DQI coding accuracy for two modern LLMs (OpenAI’s GPT 4.1 Nano and Google’s Gemini 2.0 Flash), systematically varying examples from zero-shot up to 50 to identify performance scaling patterns and optimal ranges (cf. Dong *et al.*, 2024). Second, I conduct a comparative analysis of performance between these models, considering associated costs, to provide practical guidance on model selection for large-scale DQI annotation (cf. Bojic *et al.*, 2025; Sun *et al.*, 2024). Third, I perform a detailed error analysis on a high-performing configuration to understand the nature of residual errors and identify specific DQI dimensions or judgment types that remain challenging for current models (cf. Li, 2025). Fourth, I develop and open-source a modular, adaptable AI engineering pipeline that enables systematic experimentation with LLM prompting across various measurement tasks, providing a technical foundation that other researchers can build upon for their own qualitative measurement challenges.

This study demonstrates that prompting strategies that combine codebook-style annotation instructions with a handful of examples of correct annotations enable LLMs to perform complex qualitative coding like the DQI at levels approaching expert human annotators significantly enhancing the feasibility of large-scale deliberative quality research. However, the results presented here suggest that understanding model-specific performance and limitations through systematic evaluation and error analysis is vital for responsible application and identifying areas for future work, such as the exploration of more advanced reasoning prompts (Dunivin, 2024).

This paper proceeds as follows. Section 2 reviews related work in computational text analysis, LLMs for qualitative coding, and the DQI. Section 3 details my data, prompt design, models, and evaluation metrics. Section 4 presents the findings from the ICL scaling, model comparison, reliability analysis, and error analysis. Section 5 discusses the implications of these findings. Section 6 concludes.

2 Previous Work

This paper intersects with research on computational text analysis, the rapidly evolving capabilities of LLMs for complex tasks, and the specific challenge of measuring deliberative quality.

2.1 Computational Methods for Measuring Textual Constructs

Early approaches to the quantitative analysis of text often relied on dictionary-based methods, counting words associated with specific concepts (Grimmer and Stewart, 2013; Tausczik and Pennebaker, 2010), though these methods face limitations in capturing context and subtlety (Ribeiro *et al.*, 2016). Supervised machine learning techniques offered improvements by training models on human-labeled data to classify texts according to predefined categories (Rudkowsky *et al.*, 2018; Card *et al.*, 2015), but require significant labeling effort and may not generalize well. Unsupervised methods like topic modeling (Blei *et al.*, 2003) and word embeddings (Mikolov *et al.*, 2013) can uncover latent structures without labeled data (Garg *et al.*, 2018; Sagi and Deignan, 2013), but interpreting these structures and mapping them to specific theoretical constructs remains challenging. These methods generally struggle to replicate the fine-grained, multi-dimensional judgments typical of expert qualitative coding schemes like the DQI.

2.2 Large Language Models and In-Context Learning for Text Analysis

Recent advances in LLMs, powered by architectures like the Transformer (Vaswani *et al.*, 2017), offer new possibilities for text analysis (Bommasani *et al.*, 2021; Brown *et al.*, 2020). These models learn rich representations of language during pre-training on vast corpora. A key feature is their ability to adapt to new tasks via prompt engineering, often without requiring task-specific fine-tuning (Liu *et al.*, 2021). In-Context Learning (ICL), where the model learns from examples provided directly in the prompt, is particularly relevant for applying LLMs to specialized tasks with limited labeled data (Brown *et al.*, 2020). Research suggests that performance often improves significantly with a small number of examples, followed by diminishing returns (Dong *et al.*, 2024; Omizo, 2024). While other techniques, such as Chain-of-Thought (CoT) reasoning, show promise for improving performance on complex tasks by prompting step-by-step logic (Wei *et al.*, 2022; Kojima *et al.*, 2022; Dunivin, 2024), this study focuses primarily on optimizing performance through the systematic variation of ICL examples.

2.3 Applying LLMs to Qualitative Coding and Political Text

There is a growing interest in applying LLMs to automate tasks traditionally performed through manual qualitative coding. Studies have demonstrated the potential of LLMs, particularly when guided by structured prompts mimicking codebooks (Xiao *et al.*, 2023), to achieve reasonable agreement with human coders on various qualitative schemes (Omizo, 2024). Within political science and communication research, LLMs are being explored as tools for analyzing

political texts (Ornstein *et al.*, 2025), measuring concepts like ideology or sentiment, and potentially automating complex measurement tasks (Halterman and Keith, 2025). Some work has begun applying LLMs to analyze debates or deliberation-like features, such as argument strength or interaction dynamics (Liang *et al.*, 2024; Behrendt *et al.*, 2024; Karanam *et al.*, 2024), and have shown promise in capturing complex, difficult-to-measure aspects of discourse.

2.4 The Discourse Quality Index: Measurement and Automation Efforts

The DQI, developed by Steenbergen *et al.* (2003), remains a cornerstone for empirically assessing deliberative quality in political communication. Rooted in Habermasian discourse ethics, it provides a theoretically grounded, multi-dimensional framework designed to operationalize key deliberative ideals into measurable indicators.

The DQI conceptualizes deliberation through several core components, assessed typically at the level of the individual speech act. First, it considers basic **Participation**, evaluating whether speakers can engage freely without interruption (coded simply as impaired or normal). Second, it focuses heavily on the **Quality of Justification**, assessing both the *level* of reasoning (ranging from mere assertions to sophisticated chains of inference) and the *content* or nature of the reasons provided (distinguishing between appeals to narrow interests, the common good, or principles like Rawlsian justice). Third, the DQI measures **Respect** shown towards others in multiple forms: respect towards specific social groups, respect towards the *demands* or proposals of opponents, and respect towards *counterarguments* themselves (each assessed on separate ordinal scales reflecting increasing levels of acknowledgment or validation). Finally, it evaluates the **Constructiveness** of the political engagement, coding whether speakers engage purely positionally or make efforts towards compromise, consensus, or mediating proposals. These dimensions are summarized in Table 1. Each dimension is measured using carefully defined ordinal scales, requiring trained coders to make judgments based on the textual content and its context.

The DQI framework has proven influential and has been widely applied across various contexts, including national parliaments (Wendler, 2013), televised debates (Rosset and Wüest, 2021), and online forums (Wisniewski, 2019), often requiring adaptation for specific settings (e.g., Bächtiger *et al.*, 2009). Despite its utility and theoretical grounding, the detailed, multi-faceted nature of the DQI makes its manual application highly resource-intensive, limiting the scope and scale of empirical deliberation studies.

Consequently, automating the DQI has been a long-standing goal. Early attempts often relied on traditional machine learning techniques with feature engineering. While achieving some success, these approaches generally faced limitations in capturing the full complexity inherent in the DQI constructs,

Table 1: Discourse quality index dimensions and scoring criteria.

Dimension	Definition	Scoring Criteria
Participation	The extent to which participants are able to engage in the debate without being constrained.	0: Participation impaired, 1: Normal participation
Justification (Level)	The extent to which arguments are supported by reasons.	0: No justification, 1: Vague assertion, 2: General justification, 3: Specific justification
Justification (Content)	The quality of the justification provided.	0: Lack of facts, 1: Oversimplification, 2: Relevant examples, 3: Balanced consideration
Respect (Groups)	The extent to which participants show respect towards other groups.	0: No respect, 1: Formal respect, 2: Substantial respect
Respect (Demands)	The extent to which participants show respect towards the demands of other groups.	0: No respect, 1: Formal respect, 2: Substantial respect, 3: Acknowledges legitimacy
Respect (Counterarguments)	The extent to which participants show respect towards counterarguments.	0: No respect, 1: Formal respect, 2: Substantial respect, 3: Acknowledges validity, 4: Engages with counterargument
Constructive Politics	The extent to which participants engage in positive sum politics and try to reach compromise solutions.	0: Positional politics, 1: Alternative proposal, 2: Consensus appeal, 3: Mediating proposal

particularly for dimensions requiring deeper semantic understanding (Rincón *et al.*, 2021; Fournier-Tombs and MacKenzie, 2021). While more recent work leverages the power of LLMs for related deliberative tasks or specific components (Liang *et al.*, 2024; Behrendt *et al.*, 2024; Karanam *et al.*, 2024), this paper provides an evaluation of modern LLMs ability to replicate the original, complete DQI framework.

3 Methodology

In this section, I detail the methodology used to evaluate the performance of LLMs in automating the coding of the DQI.

3.1 Data

I use a validation dataset derived from Steenbergen *et al.* (2003), consisting of 1044 parliamentary speeches from the 101st and 104th US Congress manually annotated by expert coders on the seven DQI dimensions (summarized in Table 1). The raw speech texts were sourced from the Congressional Record corpus (Gentzkow and Shapiro, 2017). Of the 1044 annotated speeches, 614 were complete, with annotations provided for each of the dimensions of the DQI. Following standard practice, these 614 speeches were split into a training set (80%, $N = 495$) used solely for selecting ICL examples, and a validation set (20%, $N = 119$) used for evaluating model performance. The validation set remained unseen by the models during the example selection process.

3.2 Prompt Design and ICL

A prompt template was developed to provide the LLM with the necessary context and instructions for performing DQI coding. This template aimed to simulate the information provided to a human coder, including a definition of the coder’s role (expert political scientist), a theoretical overview of the DQI, detailed coding criteria derived directly from the original DQI manual (Steenbergen *et al.*, 2003), a structured JSON output format specification (Figure 4), and the target speech text. A conceptual overview of the prompt structure is shown in Figure 1. For an example of a complete prompt, see the Online Appendix.

3.2.1 Theoretical Foundation

The prompt provides an overview of the DQI framework and its theoretical foundations in the context of deliberative democracy. See Figure 2 for an excerpt of the theoretical foundation provided to the model. This formulation

```

You are an expert political scientist. Analyze the given
parliamentary speech and annotate it on the following dimensions
of the Discourse Quality Index (DQI).

Theoretical overview of the DQI:
{dqi-theory}

Annotation criteria for each dimension:
{dqi-criteria}

Structure annotations in the following format:
{output-format-json-schema}

Examples of annotated speeches:
{example_annotation-1}
...
{example_annotation-n} (*Varied from 0 to 50*)

Annotate the following speech:
{target-speech-text}

Final instructions:
{final-instructions}.

```

Figure 1: Conceptual overview of the prompt template structure provided to the Large Language Models.

was taken directly from the appendix in Steenbergen *et al.*, 2003 that provides the theoretical foundation of the DQI.

3.2.2 Dimension Definitions and Scoring Criteria

The prompt also details the dimensions and scoring criteria of the DQI, providing descriptions and examples of the scoring levels for each dimension and guiding the model to consider evidence for each possible score before making a final decision. See Figure 3 for an excerpt of the dimension definitions and scoring criteria provided to the model. As with the theoretical foundation, this information was taken directly from the appendix in Steenbergen *et al.*, 2003 that provides the dimension definitions and scoring criteria of the DQI.

3.2.3 Output Format

The prompt establishes a structured output format for the annotations, including fields for each dimension and sub-dimension, as well as additional fields

The DQI attempts to measure political deliberation in a general, valid and reliable way (Steenbergen et al. 2003). It mainly draws on Habermasian discourse ethics, but also incorporates elements of other deliberative models. The unit of analysis of the DQI is a speech act, i.e. the discourse by a particular individual delivered at a particular point in a debate. For each speech, we distinguish between relevant and irrelevant parts, and only the relevant parts are coded. A relevant part is one that contains a demand, i.e. a proposal on what decision should or should not be made. Our emphasis on demands stems from the fact that they constitute the heart of the deliberation. That is, demands stipulate what ought to be and what ought not to be, and this normative character puts them at the center of discourse ethics. The DQI is composed of seven indicators. Despite the considerable complexity of parliamentary debates, we attempt to keep the coding categories relatively easy, so as to ensure a high level of reliability. The following is an elaboration of these indicators, followed by an overview of the indicators and their codes. We discuss seven the seven indicators under four headings.

1. Participation. Participation constitutes a fundamental precondition for deliberation. In parliamentary settings of western democracies, this type of basic participation can usually be seen as given for the elected representatives. Normal participation is only assumed to be impaired if a speaker is cut off by a formal decision, or if she or he feels explicitly disturbed in the case of a verbal interruption by other actors.

...

Figure 2: Excerpt of DQI theoretical foundation provided to the model.

for overall notes and summary if specified. See Figure 4 for the JSON¹ output format provided to the model.

3.2.4 Sample Speeches and Annotations

The prompt includes a set of sample speeches with corresponding annotations, with the goal of both establishing the expected level of detail and reasoning in the evidence/score justifications and providing a model for the annotator

¹JSON (JavaScript Object Notation) is a lightweight data-interchange format that is easy for humans to read and write and easy for machines to parse and generate.

```

1. Participation
0: participation was impaired - speaker was cut off or
  explicitly disturbed
1: normal participation was possible

2. Justification
2.1 Level of Justification
0: no justification
1: inferior justification - reason given but not properly
  linked to demand
2: qualified justification - one complete linkage between
  demand and reason
3: sophisticated justification (broad) - multiple complete
  justifications
4: sophisticated justification (in depth) - multiple
  justifications with one embedded in complete inference
  chain

2.2 Content of Justification
0: explicit reference to group/constituency interests
1: neutral statement, no explicit references to group
  interests
2: explicit reference to common good (utilitarian/collective)
3: explicit reference to helping least advantaged (difference
  principle)

...

```

Figure 3: Excerpt of DQI dimension definitions and scoring criteria provided to the model.

to follow when coding the target speech. To generate these examples, I select a random subset of speeches from the validation dataset and for each speech provide the complete speech text along with the corresponding DQI annotations. As I discuss in the next section, I experiment with different numbers of randomly selected examples to determine how responsive the model is to demonstration of the desired annotation behavior.

3.2.5 Target Speech

The prompt also presents the target speech to be annotated. The target speech is the main focus of the annotation task, and the model is directed to provide detailed annotations for each dimension of the DQI based on the content of the speech. I provide the first 10000 characters of the speech to ensure that the model has sufficient context to make informed judgments.

```

```json
{
 "participation": {"score": [0/1]},
 "justification": {
 "level": {"score": [0-4]},
 "content": {"score": [0-3]}
 },
 "respect": {
 "groups": {"score": [0-2]},
 "demand": {"score": [0-3]},
 "counterargument": {"score": [0-4]}
 },
 "constructive_politics": {"score": [0-3]}
}
```

```

Figure 4: Structured JSON output format required from the LLMs for DQI annotations.

3.2.6 In-Context Learning

The core experimental manipulation focused on the effect of ICL example quantity. I generated prompt variants containing systematically varied numbers of examples, ranging from no examples (zero shot) to 50 examples in increments of 5. Each example consisted of a speech text paired with its corresponding expert-coded JSON annotation, drawn randomly without replacement from the 800-speech training set for each annotation task performed on the validation set speeches. To mitigate the influence of idiosyncratic features arising from any single random draw of examples, a Monte Carlo simulation approach was employed. Specifically, for each model and for each ICL configuration containing more than zero examples (i.e., 5 through 50 examples), the entire process of randomly selecting examples and evaluating the model on the full validation set was repeated five independent times. The resulting performance metrics (e.g., MAE, Krippendorff’s Alpha) for each model at each ICL level greater than zero were then averaged across these five draws to provide a more robust estimate of performance.

3.3 Model Selection and Parameter Configuration

This study primarily evaluates and compares two contemporary LLMs: GPT 4.1 Nano (provided by OpenAI) and Gemini 2.0 Flash (provided by Google). These models were selected because, as of this writing (April 2025), they represent the state-of-the-art in terms of capabilities-to-cost ratio for general-purpose language understanding. Both models are widely deployed in production applications and provide an optimal balance between performance and affordability,

making them particularly suitable for resource-constrained research contexts that require processing substantial volumes of text.

The selection of these specific models was also driven by pragmatic considerations. The OpenAI and Google APIs offer high throughput and reliability, enabling efficient testing across multiple experimental conditions. While I acknowledge the concerns regarding closed-source models and the limitations this places on full scientific replicability, testing across the full spectrum of possible models (including open-source alternatives) falls outside the scope of this study. Instead, I seek to determine whether affordable, modern LLMs can effectively perform complex qualitative coding tasks like the DQI with minimal fine-tuning.

For all API calls to these models, a temperature parameter of 0.7 was used to balance creativity and predictability in the generated annotations, and the maximum number of output tokens was set to 4000 to accommodate the structured JSON response. Other model parameters were left at their default API settings.

3.4 Evaluation Metrics

Model performance is primarily evaluated by comparing the LLM’s predicted DQI scores ($\hat{y}$) against the expert ground truth scores (y) from the validation set ($N = 119$ speeches). Two key metrics are employed.

First, Mean Absolute Error (MAE) is calculated, averaged across all $M = 7$ DQI dimensions. MAE provides an intuitive measure of the average magnitude of error on the original DQI scales and is formally defined here as:

$$\text{MAE} = \frac{1}{N \times M} \sum_{i=1}^N \sum_{j=1}^M |\hat{y}_{ij} - y_{ij}|$$

Lower MAE values indicate better performance. Considering typical human inter-rater reliability for DQI (e.g., Neblo *et al.*, 2018; Steiner *et al.*, 2004, $\alpha = 0.7$ – 0.8), an average MAE below 1.0 suggests performance approaching human consistency levels.²

Second, Krippendorff’s Alpha (Krippendorff, 2018) is calculated to assess inter-rater reliability between the LLM and the human gold standard. Alpha

²While MAE measures average error magnitude and Krippendorff’s Alpha measures chance-corrected agreement, they provide complementary views on performance. Human coders achieving acceptable reliability (e.g., $\alpha = 0.7$ – 0.8) on ordinal scales like the DQI (typically ranging 0–3 or 0–4) inevitably exhibit some disagreement. An MAE represents the average absolute difference between two sets of ratings. If an LLM achieves an MAE substantially below 1.0 compared to the human gold standard, it implies its average deviation is likely within the typical range of point differences observed between two reasonably consistent human coders. Thus, $4\text{MAE} < 1.0$ serves as a useful, albeit heuristic, benchmark suggesting the LLM’s error magnitude is comparable to expected human variation.

was computed separately for each DQI dimension using the `ordinal` level of measurement setting in the `krippendorff` Python package. This accounts for the magnitude of disagreements on the ordinal scales and allows for direct comparison to reliability benchmarks commonly reported in DQI research and LLM annotation studies (cf. Bojic *et al.*, 2025). Higher Alpha values (approaching 1.0) indicate better agreement than expected by chance.³

3.5 Pipeline Architecture

To evaluate the performance of LLMs on the DQI coding task across different ICL configurations, I developed a modular pipeline architecture that automates the end-to-end experimentation process. The pipeline consists of five key stages: (1) data preparation, (2) prompt generation, (3) LLM API interaction, (4) response parsing, and (5) evaluation and visualization. Each stage is implemented as a separate module in Python, enabling flexible configuration and independent execution. This approach ensures reproducibility by maintaining consistent experimental protocols and recording all generated artifacts.⁴

3.6 Error Analysis

To gain deeper insights into model limitations, a detailed error analysis was conducted on the predictions generated by a high-performing configuration (Gemini 2.0 Flash with 50 ICL examples, initialization 0). This analysis involved several steps.

First, I examine MAE and Alpha scores per DQI dimension to identify which aspects of deliberative quality were most challenging for the target LLM code reliably and generate confusion matrices for each dimension to visualize patterns in score disagreements (e.g., frequent confusion between adjacent scores).

Second, I perform a systematic qualitative review, assisted by a separate, powerful LLM (Gemini 2.5 Pro). Specifically, Gemini 2.5 Pro was employed to conduct an initial, detailed comparison between the Gemini 2.0 Flash predictions and the expert human gold standard codes for a sample of 30 validation

³For two coders, Krippendorff’s α is

$$\alpha = 1 - \frac{D_o}{D_e}, \quad D_o = \frac{1}{n} \sum_{i < j} n_{ij} \delta_{ij} \quad \text{and} \quad D_e = \frac{1}{n(n-1)} \sum_c \nu_c(n - \nu_c) \bar{\delta}c,$$

where n is the number of items, n_{ij} the frequency of the (*coder 1*, *coder 2*) category pair (i, j), ν_c the marginal frequency of category c , and δ_{ij} the ordinal-level distance between categories i and j . For ordinal data δ_{ij} is the squared difference of cumulative relative frequencies: $\delta_{ij} = (F_i - F_j)^2$, with $F_k = \sum_{m \leq k} \nu_m / n$. D_o is the observed disagreement, D_e the disagreement expected by chance; $\alpha = 1$ when $D_o = 0$ and $\alpha = 0$ when $D_o = D_e$.

⁴A complete implementation is available for researchers to use, adapt, and extend at <https://github.com/mbosley/mbosley.github.io>.

set speeches. For each speech in this sample, Gemini 2.5 Pro identified the dimensions of disagreement and provided a preliminary qualitative analysis of the potential reasons for the discrepancies. The full output of this LLM-assisted comparison is documented in the Online Appendix.

I synthesized the findings from this LLM-assisted review, interpreting the detailed comparisons from the Online Appendix to identify recurring error patterns, challenging linguistic features, and key areas where the target model struggled (cf. Li, 2025). The summary of these synthesized error patterns is presented in the main text. This collaborative approach leverages the analysis capabilities of a powerful LLM for the detailed comparison task, while retaining human oversight and interpretation for identifying the broader themes and implications of the errors.

4 Results

This section presents the results of the experiments evaluating the performance of GPT 4.1 Nano and Gemini 2.0 Flash on the DQI coding task across different numbers of In-Context Learning (ICL) examples. Performance is assessed primarily using MAE and Krippendorff’s Alpha.

4.1 Impact of In-Context Learning Examples on Overall Performance

Figure 5 illustrates the average MAE across all seven DQI dimensions as a function of the number of ICL examples provided in the prompt. Both models demonstrate a substantial improvement over their zero-shot performance when provided with even a small number of examples. The Gemini 2.0 Flash model starts with a lower zero-shot MAE (approximately 0.61) compared to GPT 4.1 Nano (approximately 0.84).

The most significant reduction in error for both models occurs between 0 and 10–15 examples. By 10 examples, both models converge to a similar MAE level of approximately 0.44. Beyond this point, their performance characteristics diverge slightly. The Gemini 2.0 Flash model’s performance stabilizes, achieving its lowest average MAE around 30–50 examples (approximately 0.40). In contrast, the GPT 4.1 Nano model exhibits more fluctuation after 10 examples, with MAE varying between roughly 0.44 and 0.54, generally settling at a slightly higher error level than Gemini 2.0 Flash. These results indicate diminishing returns for additional ICL examples beyond approximately 15–20 examples, suggesting that providing a moderate number of examples is sufficient to capture most of the achievable performance gains for this task, consistent with general findings on ICL scaling (Dong *et al.*, 2024; Omizo, 2024).

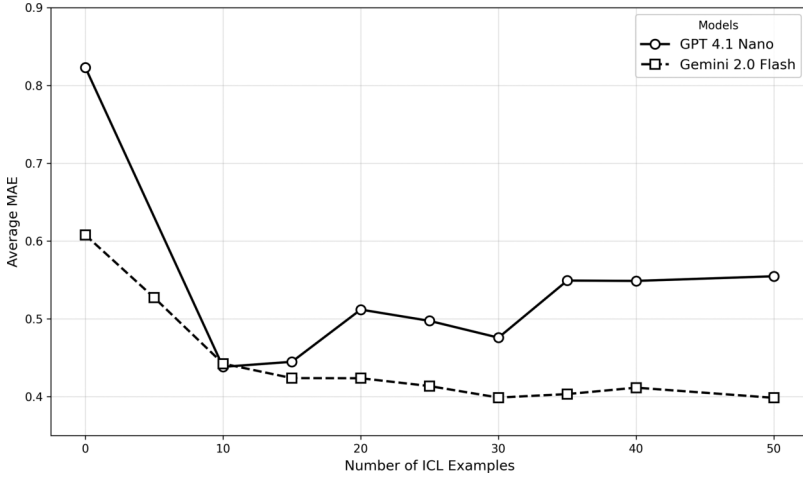


Figure 5: Overall Mean Absolute Error (MAE) across all DQI dimensions versus the number of In-Context Learning (ICL) examples provided in the prompt, shown for GPT 4.1 Nano and Gemini 2.0 Flash. Points represent average MAE over $N = 119$ validation speeches, averaged over 5 draws for $ICL > 0$.

4.2 Dimension-Specific Performance and Model Comparison

Performance varies considerably across the individual DQI dimensions, as shown in Figure 6, with some dimensions proving much easier for the models to learn than others. “Participation” was coded with near-perfect accuracy (MAE close to 0) by both models even without any ICL examples. “Constructive Politics” also showed strong performance, with both models achieving low MAE (often below 0.2) after approximately 15 examples; Gemini 2.0 Flash consistently achieved slightly lower error on this dimension in the 15–50 example range.

Conversely, certain dimensions remained challenging. “Justification-Level” exhibited persistently high MAE (above 0.8 for GPT 4.1 Nano beyond 15 examples, and above 0.6 for Gemini 2.0 Flash). Notably, Gemini 2.0 Flash consistently outperformed GPT 4.1 Nano on this dimension, and the latter model’s performance actually degraded slightly as more examples were added beyond 15. “Respect-Counterarg” also proved difficult, with MAE remaining relatively high (around 0.8–1.0) for both models, showing only modest improvement with ICL.

Performance on intermediate dimensions varied. For “Respect-Group”, Gemini 2.0 Flash achieved substantially lower MAE (around 0.1–0.15) than GPT 4.1 Nano after 10 examples. On “Respect for Demands”, both models improved significantly with ICL, reaching comparable performance levels with MAE plateauing around 0.4–0.5. For “Justification-Content”, both models

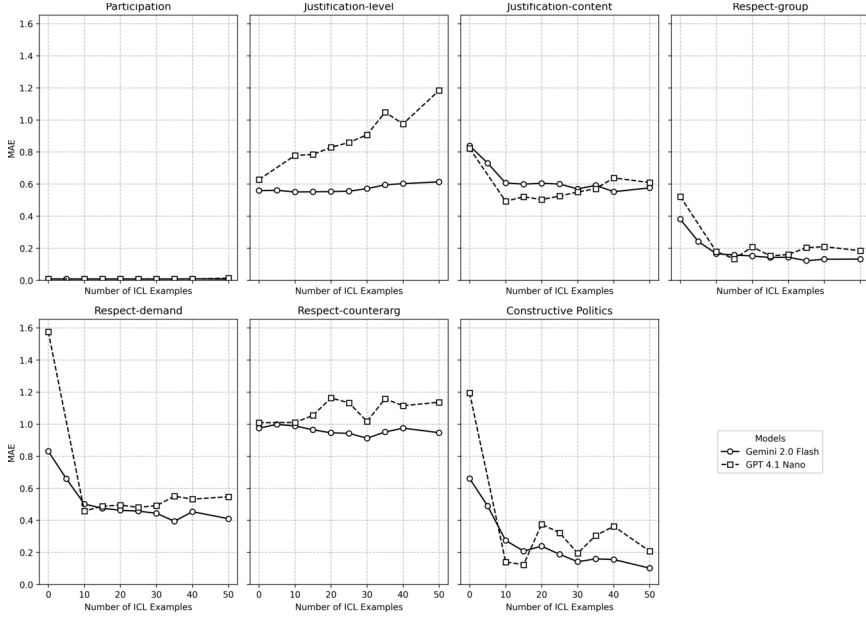


Figure 6: Mean Absolute Error (MAE) versus the number of In-Context Learning (ICL) examples, faceted by each of the seven DQI dimensions. Lines show performance trends for GPT 4.1 Nano and Gemini 2.0 Flash on each specific dimension, averaged over 5 draws for ICL > 0.

reduced MAE considerably with examples but stabilized at a moderate error level (around 0.5–0.6).

Across dimensions, Gemini 2.0 Flash generally displayed more stable error rates after the initial rapid improvement phase (0–15 examples). It demonstrated a clear advantage on “Justification-Level” and “Respect-Group” compared to GPT 4.1 Nano in the few-shot setting.

4.3 Reliability Analysis

To assess the reliability of the LLM annotations against the human gold standard, Krippendorff’s Alpha (ordinal) was calculated for each dimension. Figure 7 presents these results across different ICL levels for both models. The Alpha scores generally mirror the MAE findings, with dimensions exhibiting low MAE achieving higher reliability scores.

Several dimensions reached levels of reliability often considered “Substantial” (α between 0.67 and 0.80) or “Excellent” ($\alpha > 0.80$) in social science research (cf. Bojic *et al.*, 2025; Krippendorff, 2018) when sufficient ICL examples (typically 10–15 or more) were provided. Specifically, “Participation” consistently

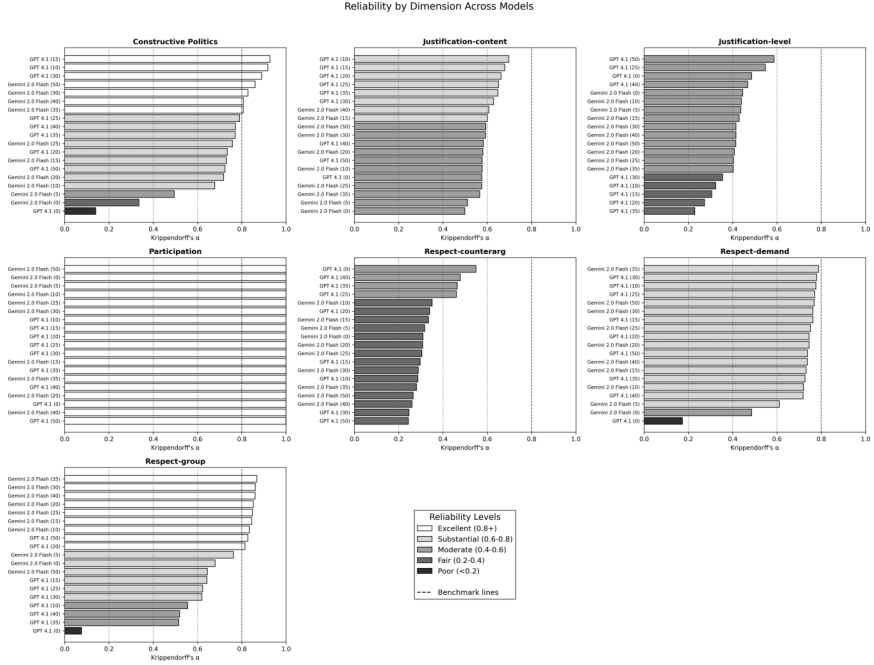


Figure 7: Krippendorff's Alpha (Ordinal) by DQI Dimension for GPT 4.1 Nano and Gemini 2.0 Flash across different numbers of ICL examples (0–50). Bars represent alpha values, averaged over 5 draws for ICL > 0. Dashed lines indicate conventional reliability benchmarks.

achieved near-perfect reliability. “Respect-Group”, “Respect-Demand”, and “Constructive Politics” frequently reached substantial or excellent agreement for both models, with Gemini 2.0 Flash often achieving higher scores or reaching thresholds with fewer examples, notably hitting excellent reliability on “Respect-Group”.

In contrast, the dimensions identified as challenging based on MAE also showed low reliability. “Justification-Level” and “Respect-Counterarg” typically exhibited Alpha scores in the “Poor” ($\alpha < 0.20$) to “Fair” ($0.20 \leq \alpha < 0.40$) ranges, indicating that the models’ ratings on these dimensions had agreement levels only slightly better, or sometimes worse, than chance compared to the human coders. “Justification-Content” generally achieved “Moderate” reliability ($0.40 \leq \alpha < 0.67$).

The positive effect of ICL examples on reliability is evident across most dimensions, with Alpha scores generally increasing and stabilizing after 10–15 examples, mirroring the MAE stabilization point. Table 2 summarizes the reliability and MAE scores for a high-performing configuration. Overall, while the models demonstrate the capacity for reliable coding on several DQI

Table 2: Performance of Gemini 2.0 Flash (50 ICL examples) on the validation set. Values are averaged over the five random-seed initializations; lower MAE and higher α indicate better agreement with expert labels.

| DQI Dimension | Krippendorff’s α | MAE |
|-----------------------|-------------------------|-------|
| Participation | 0.9996 | 0.008 |
| Justification–Level | 0.4143 | 0.614 |
| Justification–Content | 0.5936 | 0.576 |
| Respect–Group | 0.6446 | 0.132 |
| Respect–Demand | 0.7683 | 0.410 |
| Respect–Counterarg | 0.2652 | 0.947 |
| Constructive Politics | 0.8597 | 0.102 |
| Average | 0.6493 | 0.398 |

dimensions comparable to human standards, significant reliability challenges persist for others.

4.4 Cost Comparison

Beyond accuracy and reliability, the practical feasibility of using these models at scale depends on their computational cost. Figure 8 presents the total cost incurred to process the entire validation set ($N = 119$ speeches) for each model and ICL configuration.

As expected, the cost increases substantially with the number of ICL examples for both models. This is because adding examples directly increases the length of the prompt sent to the model, thereby increasing the number of input tokens processed per API call. The cost ranges from approximately \$0.03 for zero-shot inference to around \$0.50 for inference with 50 examples, representing a roughly 15–17 fold increase across this range.

The performance analyses (Figures 5 and 6) showed that performance gains diminish significantly beyond 15–20 examples, with Gemini 2.0 Flash achieving near-optimal and stable performance around 30 examples (MAE = 0.40). However, the cost continues to rise steeply beyond this point. Using 50 examples instead of 30 increases the cost for Gemini 2.0 Flash by nearly 90% (from \$0.257 to \$0.487) for virtually no improvement in average MAE. Similarly, for GPT 4.1 Nano, increasing examples from its performance plateau around 10–15 examples up to 50 leads to a large cost increase (from \$0.17 to \$0.525) without a clear, sustained improvement in average performance.

These findings highlight a clear cost-performance trade-off. While adding more examples up to a point is essential for accuracy, exceeding the saturation point (15–30 examples for this task) dramatically increases costs with little to no corresponding benefit in overall coding quality. Furthermore, considering

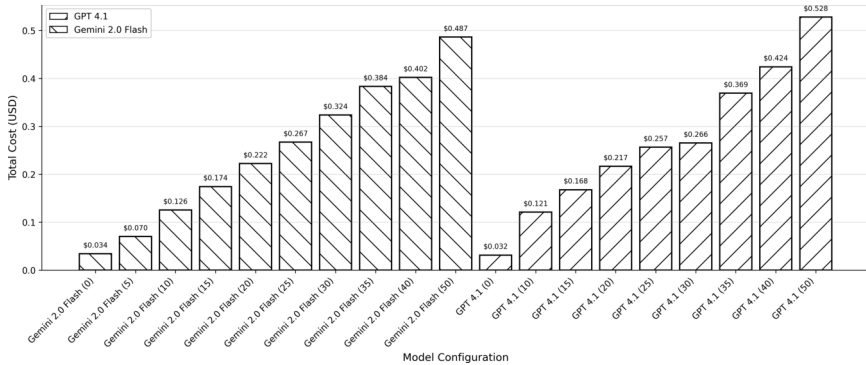


Figure 8: Total Cost (USD) to process the validation set ($N = 119$ speeches) for GPT 4.1 Nano and Gemini 2.0 Flash across different numbers of ICL examples (0–50).

both performance and cost, Gemini 2.0 Flash appears generally more cost-effective in the few-shot regime for this specific DQI coding task, offering slightly better and more stable accuracy at a lower computational expense compared to GPT 4.1 Nano when using 5 or more examples.

4.5 Error Analysis

While the aggregate performance metrics demonstrate considerable success, particularly with sufficient ICL examples, a detailed qualitative analysis focusing on the Gemini 2.0 Flash model (50 ICL examples) provides deeper insights into the model’s coding behavior and reveals specific, recurring patterns that explain the observed performance variations across dimensions. Examination of confusion matrices (Figure 9) alongside qualitative review of individual speech discrepancies (drawing from the sample analyzed in the Online Appendix) highlights these tendencies.

The analysis confirms the model’s strengths on dimensions with clear, often surface-level cues. As indicated by the near-perfect confusion matrix for “Participation” (Figure 9a), performance was nearly flawless, likely because the model can easily identify the absence of explicit interruptions mentioned in the transcript. Similarly, dimensions achieving high reliability like “Constructive Politics” (Figure 9g) and “Respect-Group” (Figure 9d) show strong diagonal concentrations in their matrices for the most frequent categories. The model reliably identifies “Positional Politics” (Level 0) for Constructive Politics (71/77 cases in the sample analyzed) and “Neutral/Implicit” respect (Level 1) for Respect-Group (74/77 cases). However, qualitative review revealed minor systematic tendencies even here. For “Constructive Politics”, there was a slight over-ascription of compromise based on procedural or detailed legislative outlining. For instance, in one speech (Speech 21 by Mr. Biden), a lengthy

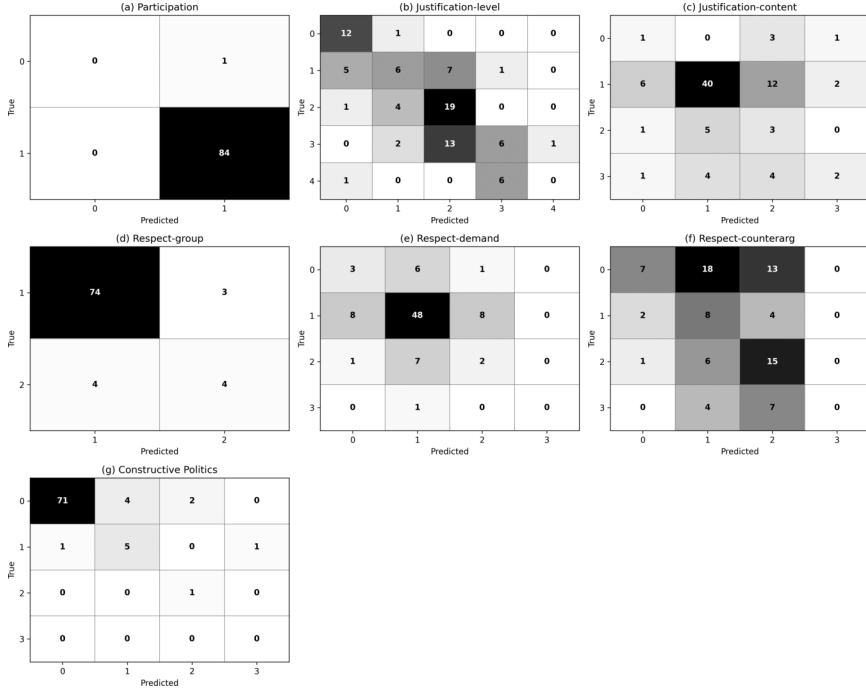


Figure 9: Confusion matrices for Gemini 2.0 Flash (50 ICL examples, init 0) on the validation set ($N = 119$). Rows denote expert labels; columns denote model predictions; darker cells indicate more items.

section detailing and comparing the features of the Democratic crime bill versus the President’s crime bill, while clearly advocating for the former, was coded by the model as a “Mediating Proposal” (Level 2), likely triggered by the detailed outlining of alternative policy frameworks, whereas human coders saw it as standard positional advocacy (Level 0). For “Respect-Group”, the model sometimes under-detected positive respect unless strong positive sentiment words were present. In Speech 15, Mr. Fazio states: “Our approval of H.R. 2 will send a strong signal that we value these workers and their role in American society.” This was coded as Level 1 (Neutral) by the model, whereas human coders recognized the explicit positive valuation of “workers” as Level 2 respect.

For dimensions showing moderate overall performance, specific error patterns explain the observed confusion. In “Justification-Content” (Figure 9c), while the model reliably identifies the frequent “Neutral” category (Level 1, 40/58 correct in sample), its difficulty differentiating value-based justifications (Levels 0, 2, 3) often stems from a heuristic focus on entity keywords. For

example, in Speech 1, Mr. Dreier quotes a letter arguing against the minimum wage:

This line of thinking runs counter to the basic principles of a free society. Government edicts do not make wages rise; they rearrange and redistribute existing wealth. Wages rise in response to the creation of new wealth through greater productivity. Government cannot create or produce anything. Mr. Speaker, we should in fact reject this. We should improve the standards of those at the lower end of the economic spectrum by decreasing the tax and regulatory burden imposed on the private sector.

Human coders identified the appeal to “basic principles of a free society” and “greater productivity” as Level 3 (universal principles/common good), but the model focused on the mention of the “private sector,” coding it as Level 1 (self/group interest). Similarly, for “Respect-Demand” (Figure 9e), where the common “Neutral/Implicit” respect category (Level 1) is relatively well-identified (48/64 correct), errors often involve misclassifying both disrespect (Level 0) and explicit respect (Level 2) as Level 1. This arises from the model’s tendency to under-detect subtle respect cues common in legislative discourse. Standard procedural statements of support, such as Mr. Alexander’s in Speech 3: “Mr. Chairman, I rise in support of the gentlemen’s amendment,” which human coders typically register as explicit respect (Level 2), were frequently coded by the model as merely neutral (Level 1), likely because it lacks overt positive sentiment beyond the procedural function.

The dimensions identified as most difficult—“Justification-Level” (Figure 9b) and “Respect-Counterarg” (Figure 9f)—exhibit the most significant off-diagonal spread in their confusion matrices, and the qualitative analysis pinpoints the reasons. For “Justification-Level”, where the model often scores one level lower than humans (note the significant number of true Level 3 cases predicted as Level 2), the bias appears driven by an over-reliance on explicit discourse markers. The model seems to look for words like “because” or “therefore” and struggles when reasoning is implied or embedded within dense examples or comparisons without clear argumentative signposting. Consequently, it frequently downgrades justifications that humans perceive as well-supported. In Speech 23, Mr. Gekas provides a detailed comparison:

For instance, the bill that is before us says that the death penalty by and large must be relegated to only those killings which come about with an intent to kill, but we say . . . when an actor . . . acts with reckless disregard . . . like a driveby killing . . . we believe that that individual should be just as subject to the possibility of the death penalty . . . That is an outstanding difference . . . Here we are again, but here is a significant difference that proves our point by

the exception that the majority put in their bill... they put in a provision that says a driveby killing... What does that mean? It means they drew it so narrowly... that if the individual... steps out of that car... he would not be subject... So the answer is to impose our standard...

Human coders identified this detailed, multi-point critique and proposal as sophisticated justification (Level 3), but the model coded it as only qualified justification (Level 2), perhaps missing the linkages between the examples and the broader argument due to the lack of simple connective phrases.

Similarly, the widespread confusion in “Respect-Counterarg” reflects challenges beyond simple adjacent-category errors. The model struggles to track prior arguments and evaluate the nature of speaker engagement with them. Its tendency to equate lack of insults with neutrality (Level 1) and overlook pragmatic politeness leads to frequent under-scoring of respect. Furthermore, it has difficulty interpreting common rhetorical structures like acknowledging an opponent’s position respectfully before refuting it. In Speech 7, Mr. Bartlett states:

Mr. Chairman, I rise in very strong opposition to the amendment offered by the gentleman from Virginia [Mr. OLIN]. I have a great deal of respect for the gentleman from Virginia (Mr. OLIN). I have a great deal of respect for those who would urge this amendment... In my Opinion, though, this amendment would In fact be harmful...

Human coders identified the explicit statements of respect as valuing the counterargument (Level 3), despite the strong opposition. The model, however, coded this as degrading the counterargument (Level 1), seemingly unable to reconcile the explicit respect with the subsequent criticism, and perhaps interpreting the standard parliamentary courtesy as insincere or overridden by the opposition. Identifying these higher levels of respectful engagement (Level 3: explicitly valuing) proved particularly challenging for the model, with no instances correctly classified as Level 3 in the analyzed sample where the human code was 3.

In summary, the error analysis highlights that while the LLM demonstrates proficiency, particularly on dimensions with clear lexical cues, its coding strategy differs from human coders on more interpretively complex aspects. It appears to rely more on shallower discourse parsing, explicit linguistic tokens, and heuristics (like entity mentions triggering content codes or procedural language indicating support/opposition), while struggling with implied reasoning, pragmatic politeness norms distinct from overt sentiment, and tracking inter-speech argumentative relationships. These systematic differences, rather than purely random errors, explain the persistent challenges on dimensions like “Justification-Level” and “Respect Counterarg”.

5 Discussion

The results presented in this study offer insights into the capabilities and optimization of LLMs for automating complex, theory-driven qualitative coding in political science, and underscore both the potential of LLMs to transform large-scale textual analysis and the importance of carefully evaluating these models' outputs.

A central finding concerns the efficacy and scaling behavior of ICL. The dramatic improvement in performance when transitioning from zero-shot to few-shot prompting confirms the power of providing demonstrations for complex, rule-based tasks like DQI coding. More importantly, the observed saturation effect, with performance plateauing around 15-30 examples, aligns with broader trends identified in ICL research (Dong *et al.*, 2024) and specific qualitative coding applications (Omizo, 2024). This saturation point has profound practical implications: it suggests that achieving high levels of accuracy does not necessitate the creation of vast, task-specific training datasets often required by traditional supervised learning. Instead, a carefully curated set of a few dozen high-quality examples can effectively guide modern LLMs to internalize complex coding criteria, significantly lowering the barrier to entry for applying these powerful models to bespoke social science measurement tasks.

The comparison between GPT 4.1 Nano and Gemini 2.0 Flash, while specific to these models and this task, showed that while Gemini 2.0 Flash generally exhibited more stable and slightly better performance, both models achieved strong results with sufficient ICL examples. This suggests that while inherent model differences exist and may confer advantages on certain sub-tasks, effective prompt engineering—particularly the provision of relevant examples—can substantially shape and improve performance across different capable models. Researchers are therefore not solely dependent on accessing the single “best” model but can potentially achieve high performance with various LLMs through careful prompt design.

Perhaps the most significant outcome is the demonstration that LLMs can achieve levels of inter-rater reliability comparable to human standards on several core DQI dimensions. Reaching Krippendorff's Alpha scores above the conventional threshold for substantial agreement ($\alpha > 0.67$) for dimensions like “Participation,” “Respect-Group,” “Respect-Demand,” and “Constructive Politics” (cf. Bojic *et al.*, 2025) is noteworthy. It indicates that, for these facets of deliberation, the LLM can replicate expert human judgments with a consistency deemed acceptable for rigorous social science research. This capability marks a clear advance over earlier automation attempts using classical machine learning, which often struggled to capture the required interpretive depth, hitting performance ceilings well below human reliability levels (Rincón *et al.*, 2021; Fournier-Tombs and MacKenzie, 2021).

However, this study also clearly maps the current frontiers and limitations. The persistent difficulties encountered with “Justification-Level” and “Respect-Counterarg” are instructive. The detailed error analysis, including the qualitative review and confusion matrix patterns, suggests these failures stem from the models’ reliance on more superficial linguistic cues and heuristics, rather than a deeper semantic or pragmatic understanding comparable to human experts. For instance, the model’s struggle with implied reasoning, its tendency to underweight pragmatic politeness cues common in legislative discourse, and its heuristic classification of justification content based on entity mentions point to specific gaps. These are not merely random errors but indicate systematic differences in how the LLM processes information compared to a trained human coder applying the DQI framework. While the models are often “close” (as indicated by many errors being between adjacent categories), reliably making these fine-grained distinctions on the most complex dimensions remains a challenge for current architectures guided solely by ICL.

These findings carry substantial implications for empirical research on deliberation and for computational social science methodology. The ability to reliably automate coding for several DQI dimensions dramatically expands the scope of possible research. Large-N analyses of parliamentary debates across time or institutions, comparative studies of online deliberation platforms, or dynamic tracking of discourse quality become significantly more feasible. This potential to analyze previously intractable volumes of text could unlock new insights into the conditions and consequences of deliberative politics. Yet, the variable performance across dimensions suggests that the annotations of generative LLMs should not be trust implicitly. Researchers must be transparent about which dimensions their automated analysis captures reliably. Drawing strong conclusions based solely on dimensions where LLM performance is currently weak (like Justification-Level) would be inappropriate. Instead, studies might focus analyses on the reliably coded dimensions, employ automated coding as a first pass to identify segments for closer human analysis, or integrate LLM outputs with human validation, particularly for the most challenging constructs.

The limitations of this study—its focus on a specific dataset from the US Congress, the evaluation of only two LLMs, and the use of one specific prompt implementation—naturally suggest avenues for future research. Testing generalizability across different political contexts, languages, and text genres is paramount. Continued evaluation of newer LLM architectures will be essential as the technology evolves. Future work should focus on targeted strategies to improve performance on the identified challenging dimensions.

6 Conclusion

This paper evaluated the use of modern LLMs (OpenAI’s GPT 4.1 Nano and Google’s Gemini 2.0 Flash), guided by ICL, to automate the coding of the DQI in parliamentary speech.

The findings confirm that LLMs can achieve high fidelity in coding several DQI dimensions, reaching levels of reliability comparable to expert human annotators when provided with a moderate number of ICL examples (typically 15-30). Performance rapidly improves from zero-shot to few-shot settings, with diminishing returns beyond this optimal range. While model choice influences performance on specific dimensions, effective ICL prompting enables strong results from different capable models. However, the study also highlights current limitations, with dimensions requiring deep assessment of justification quality and counterargument engagement remaining challenging for the models, as evidenced by lower reliability scores and specific error patterns.

This research supports the argument that prompt-engineered LLMs offer a powerful tool for significantly enhancing the scale and efficiency of complex qualitative measurement in the social sciences. By carefully optimizing prompting strategies, particularly the number of ICL examples, researchers can automate substantial portions of labor-intensive coding tasks like the DQI, enabling new avenues for large-scale empirical inquiry into phenomena like deliberative democracy. The open-source pipeline architecture developed for this study provides a practical framework for researchers to conduct their own systematic prompting experiments, further advancing the application of AI to measurement challenges in social science research.

Future work should focus on addressing the identified limitations. A promising direction involves exploring advanced prompting techniques, such as Chain-of-Thought reasoning (Dunivin, 2024; Wei *et al.*, 2022), specifically targeted at improving performance on the dimensions where models currently struggle (Justification-Level, Respect-Counterarg). Additionally, investigating whether more computationally intensive “thinking-optimized” models, such as OpenAI’s o-series, might yield better results on these complex dimensions represents a particularly intriguing avenue. These models, which trade efficiency for deeper reasoning capabilities, might better handle the contextual interpretation required for dimensions that exhibit persistent challenges with current models, potentially echoing the inference-time scaling results observed in computer science research (Brown *et al.*, 2024). The pipeline architecture developed for this study can be readily adapted to investigate such questions with minimal modifications.

Beyond model exploration, investigating the potential benefits of model fine-tuning (Halterman and Keith, 2025) or using ensemble methods combining multiple model judgments (Behrendt *et al.*, 2024) also warrants attention. Expanding the evaluation to more diverse datasets, institutional contexts,

and languages is crucial for assessing the generalizability of these findings. Ultimately, developing robust human-in-the-loop systems, where LLMs handle high-reliability coding and flag difficult cases for human review, may offer the most effective balance between scale, efficiency, and accuracy for deploying these powerful tools in social science research.

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Appendix to: Towards Qualitative Measurement at Scale

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April 25, 2025

A Discourse Quality Index (DQI) Annotation Prompt

This appendix provides an example of a complete prompt that would be provided to a generative AI model via API. It is formatted for readability here, but would be presented as a single plain text string in practice.

A.1 Instructions

You are an expert political scientist. Analyze the given parliamentary speech and annotate it on the following dimensions of the Discourse Quality Index (DQI), providing your scores in the specified JSON format.

A.2 DQI Theoretical Context

The DQI attempts to measure political deliberation in a general, valid and reliable way (Steenbergen et al. 2003). It mainly draws on Habermasian discourse ethics, but also incorporates elements of other deliberative models. The unit of analysis of the DQI is a speech act, i.e. the discourse by a particular individual delivered at a particular point in a debate. For each speech, we distinguish between relevant and irrelevant parts, and only the relevant parts are coded. A relevant part is one that contains a demand, i.e. a proposal on what decision should or should not be made. Our emphasis on demands stems from the fact that they constitute the heart of the deliberation. That is, demands stipulate what ought to be and what ought not to be, and this normative character puts them at the center of discourse ethics. The DQI is composed of seven indicators. Despite the considerable complexity of parliamentary debates, we attempt to keep the coding categories relatively easy, so as to ensure a high level of reliability. The following is an elaboration of these indicators, followed by an overview of the indicators and their codes. We discuss seven the seven indicators under four headings.

A.2.1 1. Participation

Participation constitutes a fundamental precondition for deliberation. In parliamentary settings of western democracies, this type of basic participation can usually be seen as given

for the elected representatives. Normal participation is only assumed to be impaired if a speaker is cut off by a formal decision, or if she or he feels explicitly disturbed in the case of a verbal interruption by other actors.

A.2.2 2. Justification

Fruitful discourse requires the justification of assertions and validity claims. That is, assertions should be introduced and critically assessed through “the orderly exchange of information and grounds between parties” (Habermas 1992: 370). The major problem here is that since the ideal speech situation itself has no content, one cannot apply external standards to what constitutes a good reason; otherwise one would also predetermine the result of a discourse. Hence, we only judge to what extent a speech gives complete justifications for demands and thus makes it accessible to rational critique. The first indicator of the concept of justification is the level of the justification, whereby we distinguish between five levels. No justification forms the lowest level of the indicator. Next comes inferior justification, i.e., a conclusion is embedded in an incomplete inference (i.e., either the premise is not clear or the inference from premise to conclusion is invalid). The next level is qualified justification, where one conclusion is embedded in a complete inference (i.e., the premise is clear and the inference is valid). At the next level, sophisticated justification (broad), more than one conclusion is embedded in complete inferences. Finally, the highest level is sophisticated justification (in depth), where one or several conclusions are embedded in several complete inferences. The second indicator of justification is its content. Here, we distinguish four levels. The lowest level is explicit statement concerning constituency or group interests. The next level is a neutral statement, which either contains no reference to constituency or group interests or references to several group interests. The third level is an explicit statement in terms of a conception of the common good in utilitarian or collective terms. The highest level is an explicit statement in terms of the difference principle (Rawls 1971).

A.2.3 3. Respect

Respect for other participants is an essential feature of deliberation. We distinguish three types of respect. The first type is respect toward the group in general, or empathy, whereby speakers make explicit positive statements concerning the group to be helped. The second type of respect is respect toward demands, whereby speakers explicitly value or agree with the demands of others. The third type of respect is respect toward counterarguments, whereby speakers explicitly value or agree with counterarguments to their own position.

A.2.4 4. Constructive politics

Constructive politics is the notion that speakers go beyond mere positional politics, where they merely reiterate their own demands. Instead, they propose alternative solutions or mediating proposals, thus making an effort to build consensus. We distinguish four levels of constructive politics. The lowest level is positional politics, where speakers sit on their positions. The next level is alternative proposals, where speakers propose a different agenda. The third level is consensus appeal, where speakers make an unspecific appeal for compromise

or consensus. The highest level is mediating proposals, where speakers make a proposal within the same agenda.

A.3 DQI Annotation Categories

1. Participation

- 0: participation was impaired - speaker was cut off or explicitly disturbed
- 1: normal participation was possible

2. Justification

(a) Level of Justification

- 0: no justification
- 1: inferior justification - reason given but not properly linked to demand
- 2: qualified justification - one complete linkage between demand and reason
- 3: sophisticated justification (broad) - multiple complete justifications
- 4: sophisticated justification (in depth) - multiple justifications with one embedded in complete inference chain

(b) Content of Justification

- 0: explicit reference to group/constituency interests
- 1: neutral statement, no explicit references to group interests
- 2: explicit reference to common good (utilitarian/collective)
- 3: explicit reference to helping least advantaged (difference principle)

3. Respect

(a) Respect toward Groups

- 0: explicitly negative statement about group to be helped
- 1: neither negative nor positive statement about group
- 2: explicitly positive statement about group

(b) Respect toward main Demand

- 0: explicitly negative statement about demand
- 1: neither negative nor positive statement about demand
- 2: explicitly positive statement about demand
- 3: agreeing with and positively valuing demand

(c) Respect toward main Counterarguments

- 0: counterarguments degraded
- 1: counterarguments ignored
- 2: counterarguments included but not valued
- 3: counterarguments explicitly valued

4: counterarguments agreed with

4. Constructive Politics

0: positional politics - merely reiterating one's position

1: alternative proposal - mediating but off current agenda

2: consensus appeal - unspecific call for compromise

3: mediating proposal - specific compromise on current agenda

A.4 JSON Schema

```
{
  "participation": {
    "score": [0/1]
  },
  "justification": {
    "level": {
      "score": [0/1/2/3/4]
    },
    "content": {
      "score": [0/1/2/3]
    }
  },
  "respect": {
    "groups": {
      "score": [0/1/2]
    },
    "demand": {
      "score": [0/1/2/3]
    },
    "counterargument": {
      "score": [0/1/2/3/4]
    }
  },
  "constructive_politics": {
    "score": [0/1/2/3]
  }
}
```

A.5 Annotation Principles

Be sure to provide any evidence that would be consistent with EACH of the possible annotation scores for each dimension. Remember that scores MUST be integers. In addition,

adhere to the following principles:

1. Use Double Quotes for Strings:

- Always use double quotes (") for string values and property names in JSON.
- Example: "name": "John Doe"

2. Escape Special Characters:

- Escape any double quotes within string values using a backslash (\).
- Example: "quote": "She said, \"Hello!\\""

3. Avoid Trailing Commas:

- Do not add a comma after the last item in an array or the last property in an object.
- Correct: {"name": "John Doe", "age": 30}
- Incorrect: {"name": "John Doe", "age": 30,}

A.6 Example Annotations

A.6.1 Example 1

[Truncated for space - speech text would be included here]

```
{
  "annotation": {
    "participation": {
      "reasoning": "Score: 1 - Normal participation was
        possible.",
      "score": 1
    },
    "justification": {
      "level": {
        "reasoning": "Score: 0 - No justification provided.",
        "score": 0
      },
      "content": {
        "reasoning": "Score: 1 - Neutral statement, no
          explicit references to group interests.",
        "score": 1
      }
    },
    "respect": {
      "groups": {
        "reasoning": "Score: 1 - Neither negative nor positive
```

```

        statement about group.",
        "score": 1
    },
    "demand": {
        "description": "Main demand in the speech",
        "reasoning": "Score: 1 - Neither negative nor positive
            statement about demand.",
        "score": 1
    },
    "counterargument": {
        "description": "Main counterargument in the speech",
        "reasoning": "Score: 2 - Counterarguments included but
            not valued.",
        "score": 2
    }
},
"constructive_politics": {
    "reasoning": "Score: 0 - Positional politics - merely
        reiterating one's position.",
    "score": 0
}
}

```

A.6.2 Example 2

[Truncated for space - speech text would be included here]

```

{
    "annotation": {
        "participation": {
            "reasoning": "Score: 1 - Normal participation was
                possible.",
            "score": 1
        },
        "justification": {
            "level": {
                "reasoning": "Score: 3 - Sophisticated justification (
                    broad) - multiple complete justifications.",
                "score": 3
            },
            "content": {
                "reasoning": "Score: 1 - Neutral statement, no

```

```

        explicit references to group interests.",
        "score": 1
    },
    "respect": {
        "groups": {
            "reasoning": "Score: 1 - Neither negative nor positive
                statement about group.",
            "score": 1
        },
        "demand": {
            "description": "Main demand in the speech",
            "reasoning": "Score: 1 - Neither negative nor positive
                statement about demand.",
            "score": 1
        },
        "counterargument": {
            "description": "Main counterargument in the speech",
            "reasoning": "Score: 0 - Counterarguments degraded.",
            "score": 0
        }
    },
    "constructive_politics": {
        "reasoning": "Score: 3 - Mediating proposal - specific
            compromise on current agenda.",
        "score": 3
    }
}

```

A.6.3 Example 3: Speech by Mr. KILDEE on Fair Labor Standards

[Truncated for space - speech text would be included here]

```

{
    "annotation": {
        "participation": {
            "reasoning": "Score: 1 - Normal participation was
                possible.",
            "score": 1
        },
        "justification": {
            "level": {

```

```

    "reasoning": "Score: 3 - Sophisticated justification (
      broad) - multiple complete justifications.",
    "score": 3
  },
  "content": {
    "reasoning": "Score: 3 - Explicit reference to helping
      least advantaged (difference principle).",
    "score": 3
  }
},
"respect": {
  "groups": {
    "reasoning": "Score: 1 - Neither negative nor positive
      statement about group.",
    "score": 1
  },
  "demand": {
    "description": "Main demand in the speech",
    "reasoning": "Score: 1 - Neither negative nor positive
      statement about demand.",
    "score": 1
  },
  "counterargument": {
    "description": "Main counterargument in the speech",
    "reasoning": "Score: 1 - Counterarguments ignored.",
    "score": 1
  }
},
"constructive_politics": {
  "reasoning": "Score: 0 - Positional politics - merely
    reiterating one's position.",
  "score": 0
}
}

```

A.6.4 Example 4: Speech by Mr. MORAN on Semiautomatic Assault Weapons

[Truncated for space - speech text would be included here]

```

{
  "annotation": {
    "participation": {

```

```

    "reasoning": "Score: 1 - Normal participation was
        possible.",
    "score": 1
},
"justification": {
    "level": {
        "reasoning": "Score: 3 - Sophisticated justification (
            broad) - multiple complete justifications.",
        "score": 3
    },
    "content": {
        "reasoning": "Score: 2 - Explicit reference to common
            good (utilitarian/collective).",
        "score": 2
    }
},
"respect": {
    "groups": {
        "reasoning": "Score: 1 - Neither negative nor positive
            statement about group.",
        "score": 1
    },
    "demand": {
        "description": "Main demand in the speech",
        "reasoning": "Score: 1 - Neither negative nor positive
            statement about demand.",
        "score": 1
    },
    "counterargument": {
        "description": "Main counterargument in the speech",
        "reasoning": "Score: 0 - Counterarguments degraded.",
        "score": 0
    }
},
"constructive_politics": {
    "reasoning": "Score: 0 - Positional politics - merely
        reiterating one's position.",
    "score": 0
}
}

```

A.6.5 Example 5: Speech by Mr. GENE GREEN of Texas

Mr. GENE GREEN of Texas: Mr. Speaker. my good friend and colleague from Wisconsin. who serves on the Committee on Economic and Educational Opportunities. would actually see an increase of 210.000 workers who would see a pay increase if we pass the Riggs amendment.

```
{
  "annotation": {
    "participation": {
      "reasoning": "Score: 1 - Normal participation was
        possible.",
      "score": 1
    },
    "justification": {
      "level": {
        "reasoning": "Score: 0 - No justification provided.",
        "score": 0
      },
      "content": {
        "reasoning": "Score: 1 - Neutral statement, no
          explicit references to group interests.",
        "score": 1
      }
    },
    "respect": {
      "groups": {
        "reasoning": "Score: 1 - Neither negative nor positive
          statement about group.",
        "score": 1
      },
      "demand": {
        "description": "Main demand in the speech",
        "reasoning": "Score: 1 - Neither negative nor positive
          statement about demand.",
        "score": 1
      },
      "counterargument": {
        "description": "Main counterargument in the speech",
        "reasoning": "Score: 0 - Counterarguments degraded.",
        "score": 0
      }
    },
    "constructive_politics": {
```

```
    "reasoning": "Score: 0 - Positional politics - merely  
        reiterating one's position.",  
    "score": 0  
  }  
}  
}
```

A.6.6 Speech to Annotate

Mr. SARBANES: Mr. President. I want to add an additional point. That is. I think many employers are supportive of an increase in the minimum wage. In fact. the employers who spoke at this press conference this morning indicated they were in favor of raising the minimum wage. One of the press people said. "If you are in favor of it. why do you not just go ahead and do it. and voluntarily raise it in your business?" This lady had an immediate comeback. right on point. She said. "If all my competitors will raise their wages. their payrolls. then I am quite prepared to do it. Otherwise. I am placed at a competitive disadvantage." In effect. under the current system. the only employer who is not responsive to the needs of his employee. in effect. dictates the standard, and it is all brought down to the lowest common denominator. For many employers. this enables them to do what they think ought to be done in any event that is. give their employees a better wage. It will be done with a level playing field in terms of competition. so that employer and I think there are not all that many if they refuse to go up. they can be at a competitive advantage against those people who are more responsive to the needs of their employee and who understand the pressures that are upon them in today's age. This. in many respects. for many employers. means they have an opportunity to do what they think ought to be done. in any event. I want to make it clear. I think there are a great many employers across the country who take that position. They are not opposed to raising the minimum wage. They recognize that by raising the minimum wage. you keep the competition on a level playing field. and therefore they support the measure that is before the Senate. I very much hope. as the Senator from New York said. when we meet tomorrow we will be able to act in a positive manner on this very important matter.

A.7 Closing Instructions

Please provide your annotation in JSON format according to the schema provided in the [JSON SCHEMA] section above. Format your JSON output with markdown-style backticks, like this:

```
```json  
{Your JSON here}
```
```

Be sure to include ALL of the relevant information from the schema provided. Respond with only the JSON output.

B Detailed Error Analysis for Sample Speeches

This appendix provides a detailed error analysis for 30 speeches from the validation set, comparing the annotations generated by `gemini-2.0-flash` (with 50 ICL examples, initialization 0) against the expert human gold standard codes. For each speech, the text is provided, followed by a table comparing the codes, and a qualitative analysis focusing on the dimensions where discrepancies occurred. As per the discussion of the error analysis method in the main text, this error analysis was conducted with the help of Google’s Gemini 2.5 Pro model, and verified extensively by the author for correctness.

B.1 Speech 1: 1de76c16c79776e4fa1ce40505d320e8

Mr. DREIER: Mr. Speaker. I want to congratulate my colleagues. on both sides of this issue. I am convinced that everyone in this House sincerely wants to see an improvement for those who are at the lower end of the economic spectrum. The other night I was snuggling up on United Airlines with my U.S. News and World Report and happened to read the letters to the editor. and saw this very thoughtful piece from Ed Grady from St. Paul Park. MN. where he says: The legislated minimum wage is a classic example of a good intention and a bad Idea. the idea being that government. by simple decree. can Increase the earning power of all marginal workers. This line of thinking runs counter to the basic principles of a free society. Government edicts do not make wages rise. they rearrange and redistribute existing wealth. Wages rise in response to the creation of new wealth through greater productivity. Government cannot create or produce anything. Mr. Speaker. we should in fact reject this. We should improve the standards of those at the lower end of the economic spectrum by decreasing the tax and regulatory burden imposed on the private sector. Let us proceed as quickly as possible with a responsible measure.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓ Match |
| Justification-level | 2 | 1 | ~ Within one |
| Justification-content | 3 | 1 | × Disagree |
| Respect-group | 1 | 1 | ✓ Match |
| Respect-demand | 0 | 1 | ~ Within one |
| Respect-counterarg | 3 | 2 | ~ Within one |
| Constructive Politics | 0 | 1 | ~ Within one |

B.1.1 Analysis:

Discrepancies arose on several dimensions requiring interpretation.

Justification-Level : Human coded 2 (Qualified), model 1 (Inferior). The model likely saw the core argument (reject minimum wage) as asserted using a quote, lacking a clear

internal premise-conclusion structure, hence 'inferior'. Human coder saw the use of the quote itself as providing a reason.

Justification-Content : Human 3 (Universal principles), model 1 (Self/group interest). The human coder focused on "basic principles of a free society." The model likely keyed in on the mention of the "private sector" later in the speech, a common heuristic error.

Respect-Demand : Human 0 (Disrespectful), model 1 (Neutral). The speaker calls the minimum wage a "bad idea" and says "reject this," which the human coder saw as disrespectful. The model might have focused on the polite opening ("congratulate my colleagues") and the indirectness of quoting, leading to a neutral classification.

Respect-Counterarg : Human 3 (Valued), model 2 (Acknowledged). The opening "congratulate my colleagues, on both sides" was likely interpreted by the human as valuing opponents (standard politeness). The model, seeing subsequent strong opposition, likely coded it as mere acknowledgment.

Constructive Politics : Human 0 (Positional), model 1 (Alternative). The model correctly identified the suggestion to decrease "tax and regulatory burden" as an alternative policy, fitting the definition of Level 1. The human coder perhaps focused only on the lack of compromise **on the minimum wage itself**.

B.2 Speech 2: 22fd706290e92f16a269e03208669623

"Mr. GRAMM: Mr. President. I did not even hear the Senator from Delaware ask to be recognized."

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|----------------|--------------|
| Participation | 1 | 1 | ✓ Match |
| Justification-level | 4 | 0 | × Disagree |
| Justification-content | 2 | 1 | ~ Within one |
| Respect-group | 2 | 1 | ~ Within one |
| Respect-demand | 2 | 1 | ~ Within one |
| Respect-counterarg | 3 | 2 ~ Within one | |
| Constructive Politics | 0 | 0 | ✓ Match |

B.3 Analysis:

This extremely short, procedural statement resulted in significant discrepancies, likely due to human coding errors or misunderstanding context not present in the snippet. The model's codings (0 for justification level/content, 1 for respect dimensions) appear much more accurate for the text provided.

Justification-Level/Content : Human coded 4 and 2 respectively, but the text offers no justification. Model correctly codes 0 and 1 (default/neutral).

Respect-Group/Demand : Human coded 2 (explicitly respectful), but the text is neutral. Model correctly codes 1.

Respect-Counterarg : Human coded 3 (valued), model 2 (acknowledged). Text does not engage counterarguments. Model’s code 2 is questionable, but 0 would be most accurate. Human code seems incorrect.

B.4 Speech 3: e8db523a6e1cad38a402d1a8a490537b

Mr. ALEXANDER: Mr. Chairman. last Christmas. five blocks from the Capitol. around the corner from where I then lived. a passenger van was riddled with machinegun bullets. The driver and occupant were killed. This movielike horror was reported to have been a fight between gangs over drugs. It was a fight between warring drug lords over illegal drug. It was also reported that the victims were criminals that were released for lack of jail space. When guns are used by criminals they become instruments of death. Guns do not kill. criminals do. All lawabiding citizens. including the hunters that I represent. want something done about the rampant crime in America. But under this bill. under these provisions. my deer rifle. my Winchester .308. when modified with a larger clip. is eligible for classification as an assault weapon which is outlawed by this bill. The problem is that criminals are roaming the streets. The Nation needs more prisons to lock up the criminals and keep them off the streets. Mr. Chairman. I rise in support of the gentlemen’s amendment.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 2 | 1 | ~ Within one |
| Justification-content | 2 | 0 | × Disagree |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 2 | 1 | ~ Within one |
| Respect-counterarg | 0 | 0 | ✓Match |
| Constructive Politics | 0 | 0 | ✓Match |

B.5 Analysis:

Justification-Level : Human 2 (Qualified), model 1 (Inferior). Speaker uses an anecdote and assertion ("Guns do not kill, criminals do") to justify supporting the amendment. Human sees this as providing a reason. Model might see the assertion as lacking clear linkage, hence 'inferior'.

Justification-Content : Human 2 (Common good - aggregate), model 0 (None). Human likely interpreted the focus on "rampant crime" and needing "more prisons" as an appeal to public safety (common good). Model might have focused on the lack of explicit "common good" framing or the mention of "hunters that I represent," perhaps confusing its classification and defaulting to 0.

Respect-Demand : Human 2 (Explicitly respectful), model 1 (Neutral). Human likely coded the concluding "I rise in support of the gentlemen's amendment" as explicit respect. Model consistently codes such procedural statements as neutral.

B.6 Speech 4: edc4d6219b63db7a71291853cffd165c

Mr. SKELTON: I thank the gentleman for yielding to me. Mr. Chairman. I compliment the gentleman from Missouri (Mr. VoLKMER) on his amendment. Should this amendment not pass, what actually will be happening is harming and prejudicing all of those lawful hunters out there who, by the way, are probably the safest people handling any type of firearms in this country. Why penalize them? I certainly think the amendment is proper, and I endorse it.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓ Match |
| Justification-level | 1 | 2 | ~ Within one |
| Justification-content | 2 | 0 | × Disagree |
| Respect-group | 2 | 2 | ✓ Match |
| Respect-demand | 1 | 2 | ~ Within one |
| Respect-counterarg | 0 | 1 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓ Match |

B.7 Analysis:

Justification-Level : Human 1 (Inferior), model 2 (Qualified). Model saw the link between hunters being "safest" and the conclusion "Why penalize them?" as a qualified justification. Human might have seen "safest" as an unsupported claim, weakening the justification to inferior.

Justification-Content : Human 2 (Common good - aggregate), model 0 (None). Human might see protecting "lawful hunters" as group interest or implicitly linked to rights/common good. Model, lacking explicit framing, defaulted to 0.

Respect-Demand : Human 1 (Neutral), model 2 (Explicitly respectful). Model correctly identified "I compliment the gentleman... I endorse it" as explicit respect (Level 2). Human coding of 1 appears incorrect here.

Respect-Counterarg : Human 0 (Ignored), model 1 (Degraded). No counterarguments are mentioned. Model's code of 1 is incorrect; it should be 0.

Okay, I will continue reformatting the error analysis for speeches 5 through 30, following the detailed, discrepancy-focused format established in the first four examples.

B.8 Speech 5: 41972e8bccd55fe5a0a974012b4466c4

Mr. STEVENS: Mr. President. I feel a little strange at this point. because the amendment that I had prepared and had been waiting to present had provisions to deal with six specific items in Senate bill 1241. The four that have just been stricken were part of the amendment. What has just been eliminated now in the amendment offered by the Senator from Delaware is: The magazine ban and registration provision. the multiple sale registration provision. the pretrial detention for gun crimes. and the misdemeanor forfeiture of rights provision. They were four of the six provisions that we had substantial objections to. The amendment I have sent to the desk provides for a mandatory instant identification of felons to prevent the purchase of guns by those who are already prohibited under existing law from purchasing guns. It is the argument of those who back the Brady bill that the Brady bill is necessary to prevent felons from getting guns. Yet. the Brady bill does not mandate any check of the history of the prospective purchasers of guns. My amendment will. It will mandate the instant criminal history background check... [Long speech detailing amendment, arguing against waiting periods with statistics, defending gun owners' rights] ... In closing. Mr. President. I would like to simply say that the gun control measures included in this act are a smokescreen... If we are serious about reducing crime I suggest we leave off discussions of these ineffectual measures and concentrate on criminals. -not honest gun owners.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 4 | 3 | ~ Within one |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 2 | 1 | ~ Within one |
| Respect-counterarg | 2 | 1 | ~ Within one |
| Constructive Politics | 1 | 1 | ✓Match |

B.8.1 Analysis:

Discrepancies arose on dimensions requiring interpretation of complexity and engagement.

Justification-Level : Human 4 (In-depth - Original Scale), model 3 (Sophisticated-broad). The speech presents multiple detailed arguments with statistics against waiting periods and for instant checks. Human saw this complexity as Level 4 (in-depth). Model recognized multiple arguments (Level 3) but perhaps didn't detect the required embedding or depth for Level 4. A close call based on interpreting the complexity threshold.

Respect-Demand : Human 2 (Explicitly respectful), model 1 (Neutral). Speaker outlines his own amendment as an alternative to the Brady Bill (opponent's demand). Human

may see engaging with the problem (preventing felons from getting guns) via an alternative proposal as respect for the demand’s underlying intent. Model likely focuses on the explicit rejection of the opponent’s specific proposal (waiting period within the Brady Bill), coding it as neutral engagement rather than explicit respect for the demand itself.

Respect-Counterarg : Human 2 (Acknowledged), model 1 (Degraded). Speaker explicitly addresses the argument backing the Brady bill ("It is the argument of those who back the Brady bill...") and counters it strongly using statistics and dismissive language ("smoke-screen," "ineffectual measures"). Human coder focused on the acknowledgment of the argument. Model likely interpreted the strong refutation and dismissive framing as degrading the counterargument.

B.9 Speech 6: 97067bfaefcb076e9557499ac734007d

Mr. VOLKMER: Mr. Chairman. I yield myself such time as I may consume. Mr. Chairman. I would first like to respond to the gentleman from Illinois . We have a communication from a John Chapman of the Killeen Police Department. He is in the second unit responding to the shooting. He says. I do not speak for the department. but I take a personal stand on the right to keep and bear arms. Although I am not a member of the National Rifle Association of America. I support NRA in its opposition to the gun control provisions of the House crime bill. Mr. Chairman. I yield such time as he may consume to the gentleman from Oregon .

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 1 | 0 | ~ Within one |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 1 | ✓Match |
| Respect-counterarg | 2 | 2 | ✓Match |
| Constructive Politics | 0 | 0 | ✓Match |

B.9.1 Analysis:

One discrepancy on the justification dimension.

Justification-Level : Human 1 (Inferior), model 0 (None). The speaker quotes a police officer expressing support for the NRA’s opposition to the crime bill’s gun control provisions. The human coder likely interpreted the use of this quote as providing a form of justification, albeit weak (inferior, Level 1). The model, seeing no explicit premise-conclusion structure from the speaker himself, determined that no justification was offered (Level 0).

B.10 Speech 7: 83bd0a3528fed94e8c0fb11799639334

Mr. BARTLETT: Mr. Chairman. I appreciate the kind and colorful words of the gentleman from New York in yielding time to me. Mr. Chairman. I rise in very strong opposition to the amendment offered by the gentleman from Virginia [Mr. OLIN]. I have a great deal of respect for the gentleman from Virginia (Mr. OLIN). I have a great deal of respect for those who would urge this amendment as an attempt to try to make this bill in their opinion much more easily understood. In my Opinion. though. this amendment would In fact be harmful both to those who are disabled and to the employers themselves... [Detailed argument against the amendment, explaining why 'undue hardship' definition is better than a monetary cap, uses examples] ...Mr. Chairman. I ask the Members to vote no on the Olin amendment. The Olin amendment attempts to set a ceiling. but in fact it sets a floor. The Olin amendment attempts to define - reasonable." but it in fact makes "reasonable" unreasonable in many cases. The Olin amendment attempts to set a Presumption. but in fact it sets a target for a lawsuit.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 4 | 3 | ~ Within one |
| Justification-content | 3 | 1 | × Disagree |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 0 | ~ Within one |
| Respect-counterarg | 3 | 1 | × Disagree |
| Constructive Politics | 0 | 0 | ✓Match |

B.10.1 Analysis:

Multiple discrepancies involving justification interpretation and respect coding.

Justification-Level : Human 4 (In-depth - Original Scale), model 3 (Sophisticated-broad). Speaker offers multiple detailed arguments against the amendment (disrupts basics, harmful to employers/disabled, floor vs ceiling, examples, comparison to 'undue hardship'). Human coder saw this depth and complexity as Level 4. Model identified multiple linked arguments, fitting Level 3, but perhaps didn't see the level of integration or careful reasoning required for Level 4.

Justification-Content : Human 3 (Universal principles), model 1 (Self/group interest). Human coder likely focused on the arguments framed around fairness to the disabled, fairness to employers, and proper legal concepts ("reasonable," "undue hardship"), interpreting these as appeals to universal principles. The model possibly focused on the explicit mentions of "disabled" and "employers" and the potential negative impacts ("harmful," "target for a lawsuit"), triggering the self/group interest classification (Level 1).

Respect-Demand : Human 1 (Neutral), model 0 (Disrespectful). Speaker states "very strong opposition" and describes the amendment using strongly negative terms ("harmful," makes "reasonable" unreasonable, "target for a lawsuit"). Model correctly identified this strong negative language directed at the demand (amendment) as disrespectful (Level 0). Human coding of neutral (Level 1) appears too lenient given the explicit negative framing.

Respect-Counterarg : Human 3 (Valued), model 1 (Degraded). Speaker explicitly states "I have a great deal of respect for the gentleman... I have a great deal of respect for those who would urge this amendment." Human coder correctly identified this explicit statement of respect as valuing the opponents/counterarguments (Level 3). Model appears to have prioritized the strong substantive disagreement and negative description of the amendment, coding the interaction as acknowledgment/degradation (Level 1). This indicates a significant model failure in capturing the subtlety of parliamentary courtesy alongside policy disagreement.

B.11 Speech 8: 96b4c71f9e3809771d1bd3b773117a51

"Mr. VOLKMER: Because last year only 743 people were killed with a rifle. a lot less than were killed even with blunt instruments or knives."

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓ Match |
| Justification-level | 1 | 0 | ~ Within one |
| Justification-content | 1 | 1 | ✓ Match |
| Respect-group | 1 | 1 | ✓ Match |
| Respect-demand | 1 | 1 | ✓ Match |
| Respect-counterarg | 0 | 2 | × Disagree |
| Constructive Politics | 0 | 0 | ✓ Match |

B.11.1 Analysis:

Discrepancies regarding justification and counterargument engagement in a very short snippet.

Justification-Level : Human 1 (Inferior), model 0 (None). The speaker provides a statistic, presumably as evidence for a preceding (unstated) claim, starting with "Because". Human coder likely saw the statistic itself as a weak (inferior, Level 1) reason. Model, seeing only the statistic and the causal connector "Because" without the claim it supports, did not recognize it as a complete justification, coding 0.

Respect-Counterarg : Human 0 (Ignored), model 2 (Acknowledged). The text snippet provides a statistic but shows no engagement with any counterarguments or opposing viewpoints. The human code of 0 (Ignored) is correct. The model's code of 2 (Acknowledged) is incorrect and likely a hallucination or misinterpretation based on the partial sentence structure ("Because...").

B.12 Speech 9: 74dc3c26ac0d660ce37307367d320699

Mr. BIDEN: Mr. President. today. the Senate votes on final passage on a crime bill that has been 2 years in the making... While it contains some provisions which I believe are misguided and many other provisions which have been oversold... it is a comprehensive package... Let us start with the Brady bill... critical aspect of our Democratic bill... Second. the assault weapon ban... key difference with the Presidents bill... Next. the death penalty... most oversold part... nonetheless... a compromise between our two bills... Like our bill. this bill bans the execution of the retarded... like the Republican bill. this bill... authorizes the death penalty even in cases where no death results. Next. local law enforcement... provision from our Democratic bill. and not the Presidents... The same is true for our increased authorization for Federal law enforcement... Next. the exclusionary rule... This bill contains our Democratic proposal. not the Presidents... Regional drug prisons and boot camps... includes our provisions. from the Democratic crime bill. Habeas corpus reform. Here. the bill now includes the habeas plan submitted by the President a terrible proposal... Next. the police corps. our bill adopts the Sasser-Specter-Graham compromise... Then we have the "alien terrorist removal act."... This terribly misguided measure... I am very pleased that this onerous provision from the Presidents crime bill is not in the crime bill... And finally. the bill... contains three provisions from our Democratic bill. not found in the Presidents bill: First. rural crime initiatives... Second. youth gang initiative... And third. drug emergency areas... These are the major provisions... Thought some come from the Presidents crime bill. the vast majority... come from our Democratic bill... Mr. President. the crime bill we are going to vote on tonight is not perfect. Some parts of it are downright awful. But the heart of its sound firearms measures... increases in police and prosecutors... these core items... will help us fight crime... So tonight. I urge my colleagues to support this crime bill...

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓ Match |
| Justification-level | 3 | 3 | ✓ Match |
| Justification-content | 2 | 1 | ~ Within one |
| Respect-group | 1 | 1 | ✓ Match |
| Respect-demand | 1 | 1 | ✓ Match |
| Respect-counterarg | 3 | 2 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓ Match |

B.12.1 Analysis:

Discrepancies centered on interpreting the basis of justification and the nature of engagement with opposing views.

Justification-Content : Human 2 (Common good - aggregate), model 1 (Self/group interest). Human coder likely interpreted the overall justification for supporting the crime bill (despite flaws) as aiming to "fight crime" and enhance public safety, fitting the common good definition (Level 2). Model might have focused heavily on the detailed comparison between the "Democratic bill" and the "President's bill," interpreting the emphasis on which party's provisions prevailed as an appeal to party/group interest (Level 1).

Respect-Counterarg : Human 3 (Valued), model 2 (Acknowledged). The speaker acknowledges provisions originating from the President's bill ("some come from the President's crime bill") and explicitly mentions compromises (death penalty). Human coder likely weighed this acknowledgment of inclusions and compromises as valuing the opponent's contribution (Level 3). Model probably focused more on the strong criticisms leveled against specific opposing provisions ("terrible proposal," "terribly misguided," "awful") and coded the overall engagement as mere acknowledgment (Level 2), not reaching the threshold for net valuation.

B.13 Speech 10: 7538ed3d21cced10fabecb0b44935285

"Mr. ANDERSON: Mr. Chairman. in my State the big cities. every one of them. are doing it and they seem pleased. I know In my own city of Long Beach they are not only doing it but they are proud of doing it."

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 1 | 1 | ✓Match |
| Justification-content | 2 | 1 | ~ Within one |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 1 | ✓Match |
| Respect-counterarg | 1 | 1 | ✓Match |
| Constructive Politics | 0 | 0 | ✓Match |

B.13.1 Analysis:

One discrepancy in justification content interpretation.

Justification-Content : Human 2 (Common good - aggregate), model 1 (Self/group interest). The speaker uses examples of cities ("my State," "my own city") seemingly implementing a policy successfully ("doing it and they seem pleased," "proud of doing it"). Human coder might infer that this implies benefits for the citizens of those cities, aligning with common good (Level 2). Model likely keyed specifically on the geographical references framed personally ("my State," "my city"), interpreting this as representing constituency or group interests (Level 1).

B.14 Speech 11: 96ecb7f1c76530f2053ede6fe5561bac

Mr. GOODLING: Mr. Chairman. I yield myself 30 seconds. Mr. Chairman. I would not want. if there is still anybody listening or looking at TV. I would not want them to believe. as a matter of fact. that we are talking about a new hire in the Ridge bill. We are talking about a first hire. I do not know who that is. Even a paper carrier. I suppose. would not qualify. because they have already been hired. The gentleman talks about reducing the increase by 10 cents. They have accelerated getting there to the point that they have gained nothing through that. so I want to make sure. and I know he would want his colleagues. at least on my side of the aisle. to understand that. Mr. Chairman. I yield 3 minutes to the gentleman from Michigan .

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 1 | 2 | ~ Within one |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 1 | ✓Match |
| Respect-counterarg | 1 | 2 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓Match |

B.14.1 Analysis:

Discrepancies on justification level and counterargument respect.

Justification-Level : Human 1 (Inferior), model 2 (Qualified). Speaker makes two points: clarifying "first hire" vs "new hire" and arguing the 10-cent reduction yields "nothing" due to acceleration. Model likely identified the second point (acceleration negating the reduction) as a specific reason linked to a conclusion (the reduction is ineffective), thus coding qualified (Level 2). Human coder might have viewed both points as assertions or corrections lacking sufficient explanatory depth, thus coding inferior (Level 1).

Respect-Counterarg : Human 1 (Degraded), model 2 (Acknowledged). Speaker addresses the opponent's point about the 10-cent reduction ("The gentleman talks about...") and directly refutes its effectiveness ("gained nothing"). Human coder interpreted this direct refutation as degrading the counterargument (Level 1). Model likely saw the engagement ("The gentleman talks about...") as acknowledgment and the refutation ("gained nothing") as substantive disagreement without explicit insult or dismissal, thus coding acknowledged (Level 2).

B.15 Speech 12: 1c0e31ac8a6052ef20a59aab28200553

Mr. HYDE: Mr. Chairman. I thank my friend for yielding to me. Mr. Chairman. in response to the distinguished gentleman from New Jersey who just spoke. for whom I have the highest admiration. he said something that I really must take

issue with. He talked about the deference on full and fair adjudication which is in my bill and certainly not in the bill that the gentleman supports. I would just like to read from a letter signed by the attorney general of the gentlemen State of New Jersey. It says: (a) a standard of federal court review which defers to full and fair adjudication by state courts and respects the integrity of state court processes. This is something we support. Importantly. the full and fair standard would not bar federal habeas review but would merely avoid federal relitigation of those issues already reasonably and fairly resolved in state court. So I do not want to emasculate this. We want to keep Federal habeas corpus viable. We want to make -it work. We want to reform it. not ruin it. and I thank the gentleman for yielding to me.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 1 | 2 | ~ Within one |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 0 | 1 | ~ Within one |
| Respect-counterarg | 2 | 0 | × Disagree |
| Constructive Politics | 0 | 0 | ✓Match |

B.15.1 Analysis:

Disagreements on justification level, respect for demand, and counterargument engagement.

Justification-Level : Human 1 (Inferior), model 2 (Qualified). Speaker uses a quote from the NJ Attorney General’s letter to support the "full and fair adjudication" standard in his bill. Model likely interpreted using this external authority (the AG letter) to bolster the argument as providing a qualified justification (Level 2). Human coder might have viewed quoting authority without adding independent reasoning or elaboration as an inferior justification (Level 1).

Respect-Demand : Human 0 (Disrespectful), model 1 (Neutral). Speaker explicitly states he "must take issue with" the opponent’s point and contrasts his bill favorably ("certainly not in the bill that the gentleman supports"). Human coder interpreted this direct disagreement and challenge as disrespectful (Level 0). Model, possibly weighing the polite opening ("highest admiration") and the absence of explicit insults, coded the overall interaction regarding the demand/position as neutral (Level 1).

Respect-Counterarg : Human 2 (Acknowledged), model 0 (Ignored). Speaker explicitly states he is responding ("in response to the distinguished gentleman...") and directly addresses the opponent’s point ("He talked about the deference..."). Human coder correctly identified this direct engagement as acknowledgment (Level 2). Model incorrectly coded 0 (Ignored), failing to recognize the explicit response and engagement with the counterargument.

B.16 Speech 13: 6dce7ee0588e46d0dd6823cb942ae238

Mr. HOYER: Mr. Chairman. I thank Chairman LUKEN for yielding me this time. I rise in strong opposition to the Lipinski amendment. The Americans With Disabilities Act Is premised on the notion that this society will no longer buy new or build new vehicles and buildings that are not accessible. Nobody. not New York. not Los Angeles. not New Jersey. nobody will buy cars which are not accessible. There are no exemptions for new cars. This amendment seeks to do that. Furthermore. this amendment seeks to retreat from todays law. The regulations now pending. issued by the administration. require commuter rails to buy accessible new cars Therefore. this not only is not consistent with the ADA protections that are being extended. it is not consistent with existIng law and the administrations pending regulations. Now. let me say that the administration does not support the LipinskiHastert amendment. They believe it is not consistent with their agreement of support for the Americans With Disabilities Act. So point No. 1. the administration is opposed to this amendment. Second. this amendment is opposed by the mayor of Chicago. with whom I talked last night.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 2 | 2 | ✓Match |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 0 | ~ Within one |
| Respect-counterarg | 0 | 2 | × Disagree |
| Constructive Politics | 0 | 0 | ✓Match |

B.16.1 Analysis:

Discrepancies regarding respect for demand and counterargument engagement.

Respect-Demand : Human 1 (Neutral), model 0 (Disrespectful). Speaker states "strong opposition" and argues the amendment "seeks to retreat from todays law" and is inconsistent with existing protections and regulations. Model correctly identified this direct opposition and negative framing ("retreat") as disrespectful towards the demand (amendment) itself (Level 0). Human coding of neutral (Level 1) appears too lenient for the explicit "strong opposition."

Respect-Counterarg : Human 0 (Ignored), model 2 (Acknowledged). Speaker argues strongly against the Lipinski amendment, citing its inconsistency with ADA premises, existing law, and administration/mayoral opposition. However, the speaker does not explicitly state or engage with Lipinski's arguments or reasons for proposing the amendment. Human code 0 (Ignored) is therefore correct. Model's code 2 (Acknowledged) is incorrect; it likely

misinterpreted arguing against the amendment as engaging with the arguments supporting it.

B.17 Speech 14: 05a1e7089340cda57300ea26bbf8634e

Mr. LOTT: Mr. President. it has taken a long time for the Senate to finally come to the point where we are today... I do wish to thank the distinguished Democratic leader for his cooperation... I also especially thank the chairman... and the ranking member of the Finance Committee... I commend them for the work they have done. I remind my colleagues today that we need to remember that small businesses play a crucial... role in the creation of new jobs... More than 75 percent of all new employment opportunities... occur in small businesses... In that context. I have always been reluctant to vote for any measure which would restrict the formation and expansion of small business. It is all too easy for Congress to promise benefits--like the increase of minimum wage and to look the other way when our legislative mandate destroys jobs... That is why I strongly support what was reported out of the Finance Committee... and why I also support very aggressively the amendment offered by Senator BOND... If we are going to impose a higher minimum wage... the least we can do is to offer some support... some buffer for small businesses... So the Bond amendment is the responsible thing to do... It would exempt... small businesses which gross less than \$500,000 a year... President Clinton's own Administrator... endorsed this concept... Senator BUMPERS proposed an even broader exemption... The Bond amendment also has a training wage... There is something worse than low wages and that is no wages... There are several other very good features in this legislation for small businesses... [Details tax relief, pension simplification, spousal IRA] ... I urge my colleagues to support that. I urge my colleagues to vote against the Kennedy amendment. There is a minimum wage increase in the Bond amendment... When you couple that minimum wage increase with these small business tax reliefs... then you have a package that really provides increased wages and protection from job loss. I urge my colleagues to vote for the Bond amendment. against the Kennedy amendment. and I yield the floor.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 1 | 2 | ~ Within one |
| Justification-content | 1 | 2 | ~ Within one |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 1 | ✓Match |
| Respect-counterarg | 0 | 1 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓Match |

B.17.1 Analysis:

Disagreements on justification level, content, and counterargument respect.

Justification-Level : Human 1 (Inferior), model 2 (Qualified). Speaker justifies supporting the Bond amendment by arguing that the minimum wage "destroys jobs" and that Bond provides a necessary "buffer" for small businesses, detailing benefits like the exemption and training wage. Model likely saw the linkage between the problem (job destruction) and the solution (Bond amendment features) as providing reasons, hence qualified (Level 2). Human coder might have viewed the "destroys jobs" claim as an unsupported assertion, weakening the overall justification to inferior (Level 1).

Justification-Content : Human 1 (Self/group interest), model 2 (Common good - aggregate). Human coder focused on the explicit emphasis on protecting and benefiting "small businesses." Model might have interpreted the arguments concerning job creation ("crucial role," "destroys jobs"), job loss protection, and training wages ("something worse... is no wages") as relating to broader economic health and employment opportunities, thus fitting common good (Level 2).

Respect-Counterarg : Human 0 (Ignored), model 1 (Degraded). Speaker explicitly urges voting against the Kennedy amendment but does not engage with the substance or arguments supporting the Kennedy amendment. Human code 0 (Ignored) is correct. Model code 1 (Degraded) is incorrect, as there's no acknowledgment, let alone degradation, of the opposing arguments.

B.18 Speech 15: c13117a3ebc38ab7e6f24cbdff98a885

Mr. FAZIO: Mr. Chairman. as a cosponsor of H.R. 2. the Fair Labor Standards Amendments of 1989. I strongly support this bill which will bring economic equity to lowwage workers. The 1938 Fair Labor Standards Act established the minimum wage as a commitment to the American belief that hard work deserves adequate compensation. It was intended to ensure that every worker would be paid enough to achieve a decent standard of living. Unfortunately. we have failed to live up to that commitment. The last time Congress approved an increase... was in 1977... Today the minimum wage remains at the 1981 level... Yet since that time. the purchasing power... has decreased by more than 30 percent... Had It kept pace with inflation. the minimum wage would now stand at \$4.57 per hour. Further. during the 1960s and 1970s. a person who worked full time at a minimum wage job could earn slightly more than the amount required to keep a family of three out of poverty. Today. however. the same worker makes only \$6.968 per year which is significantly below the 1988 poverty line... While all lowwage workers are affected. I am especially alarmed at the impact that the lowminimum wage has on women and families headed by women... women comprise more than 65 percent of those earning the minimum wage... As a matter of economic justice. it is time that we raise the minimum wage. Our approval of H.R. 2 will send a strong signal that we value these workers and their role in American society. I urge my colleagues to support H.R. 2.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 2 | 2 | ✓Match |
| Justification-content | 3 | 2 | ~ Within one |
| Respect-group | 2 | 1 | ~ Within one |
| Respect-demand | 1 | 1 | ✓Match |
| Respect-counterarg | 0 | 1 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓Match |

B.18.1 Analysis:

Disagreements on justification content, respect for group, and counterargument respect.

Justification-Content : Human 3 (Universal principles), model 2 (Common good - aggregate). Human coder likely focused on the framing using terms like "economic equity," "commitment," "decent standard of living," and "economic justice," interpreting these as appeals to universal principles (Level 3). Model might have focused more on the statistical arguments about purchasing power, inflation, poverty lines, and the impact on specific demographics (women), seeing these as relating to the aggregate well-being of segments of society (Level 2).

Respect-Group : Human 2 (Explicitly respectful), model 1 (Neutral). The speaker explicitly states, "Our approval of H.R. 2 will send a strong signal that we value these workers and their role in American society." Human coder correctly identified this statement as explicitly respectful towards the group ("lowwage workers") (Level 2). Model failed to recognize this explicit valuation, coding only neutral references (Level 1).

Respect-Counterarg : Human 0 (Ignored), model 1 (Degraded). The speech argues strongly for raising the minimum wage but does not mention or engage with any arguments against doing so. Human code 0 (Ignored) is correct. Model code 1 (Degraded) is incorrect.

B.19 Speech 16: d3903a6e97816a1eee6fbedf78210e99

Mr. BENNETT: Mr. Speaker. I rise in support of overriding the veto. I congratulate the committee. Mr. Speaker. I rise In support of this proposed minimum wage. In opposition by others. it has been urged that. there are better ways to help the poor. I see that as a tactic to defeat this meritorious legislation. If there are other things that can be done to help those of low Income those things should also be done. and not used as an excuse not to pass this legislation. I believe we should override the veto and then do whatever else also needs to be done. This is a modest proposal and we ought to pass It.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 1 | 1 | ✓Match |
| Justification-content | 1 | 2 | ~ Within one |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 2 | ~ Within one |
| Respect-counterarg | 1 | 1 | ✓Match |
| Constructive Politics | 0 | 0 | ✓Match |

B.19.1 Analysis:

Discrepancies on justification content and respect for demand.

Justification-Content : Human 1 (Self/group interest), model 2 (Common good - aggregate). Speaker supports the minimum wage explicitly to "help the poor" and "help those of low Income." Model correctly identifies this focus on aiding a disadvantaged segment of society as an appeal to common good/aggregate interests (Level 2). Human coding of 1 (Self/group interest) seems incorrect, as the appeal is framed broadly towards helping the poor, not a specific interest group the speaker belongs to or directly represents in a narrow sense.

Respect-Demand : Human 1 (Neutral), model 2 (Explicitly respectful). Speaker states "I rise In support of this proposed minimum wage" and calls the legislation "meritorious." Model correctly identifies these explicit statements of support and positive valuation as explicitly respectful towards the demand (Level 2). Human coding of 1 (Neutral) seems incorrect given the explicit positive framing.

B.20 Speech 17: 0637a6815945eed8bd0bf9c94083f5f6

Mr. MAZZOLI: Mr. Chairman. I rise in opposition to the Volkmer amendment. I want to commend the courage of my friend from New York and my friend from New Jersey who began this struggle many years ago A little over 2 years ago my hometown of Louisville. Kentucky and my congressional district experienced very nearly the same kind of horror and carnage as was experienced at Killeen. TX yesterday. We. therefore. extend our great sympathies to the people of Killeen and to. its Representative.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 1 | 0 | ~ Within one |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 2 | ~ Within one |
| Respect-demand | 1 | 0 | ~ Within one |
| Respect-counterarg | 0 | 1 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓Match |

B.20.1 Analysis:

Multiple discrepancies across justification and respect dimensions.

Justification-Level : Human 1 (Inferior), model 0 (None). Speaker states opposition to the amendment and relates a local tragedy, implying a connection. Human coder might have interpreted this implied link (shared experience of violence justifying opposition to the amendment) as a weak (inferior, Level 1) justification. Model, seeing no explicit reason given for the opposition, coded 0 (None).

Respect-Group : Human 1 (Neutral), model 2 (Explicitly respectful). Model likely interpreted the statement "extend our great sympathies to the people of Killeen" as an explicit expression of respect/empathy towards a group (Level 2). Human coder possibly focused on the groups involved in the policy debate (e.g., supporters/opponents of the amendment) where references were neutral, thus coding Level 1. This reflects a reasonable difference in interpreting the scope of "group."

Respect-Demand : Human 1 (Neutral), model 0 (Disrespectful). Speaker states "I rise in opposition to the Volkmer amendment." Model correctly codes explicit opposition as disrespectful towards the demand itself (Level 0). Human code of 1 (Neutral) seems incorrect for direct opposition.

Respect-Counterarg : Human 0 (Ignored), model 1 (Degraded). The speaker opposes the Volkmer amendment but does not engage with any arguments for it. Human code 0 (Ignored) is correct. Model code 1 (Degraded) is incorrect.

B.21 Speech 18: bcf317526f92b67afef7c99fdd6805de

Mr. DELAY: Mr. Speaker. I never thought I would see this day. but I rise in opposition to this amendment offered by my good friend. the gentleman from California . Increasing the minimum wage makes minimum sense. As a former small businessowner. I remember well what intrusive government mandates did to my business. It hurt the bottom line. it hurt productivity. it hurt competitiveness. and more important. it hurt my ability to create jobs... that is what

my Democrat colleagues refuse to talk about. They will not talk about the opportunities lost... the jobs that are not created... those people who cannot get off welfare... The Democrat Party is to job creation what Dr. Kevorkian is to health care. a jobkiller cloaked in kindness... My colleagues on the other side of the aisle have made this debate an argument of fairness... Is it fair to kill the opportunities of people who want to work...? Is it fair to kill jobs...? Is it fair to make job creation too expensive...? That is the kind of fairness that liberal Democrats conveniently ignore... When Democrats ran the Congress... not once did they talk about raising the minimum wage... they have seized on this issue as their saving grace. This saving grace for the Democrats is a coup de grace for thousands of entrylevel jobs... increasing the minimum wage is the wrong way... It is a political throwaway which will do away with thousands of jobs. For that reason. I urge my colleagues to vote against it.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 2 | 2 | ✓Match |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 0 | 0 | ✓Match |
| Respect-counterarg | 1 | 0 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓Match |

B.21.1 Analysis:

One discrepancy regarding counterargument engagement.

Respect-Counterarg : Human 1 (Degraded), model 0 (Ignored). Speaker explicitly addresses the opponent's framing ("My colleagues... have made this debate an argument of fairness...") and proceeds to strongly refute and mock this framing ("Is it fair to kill...? ... That is the kind of fairness that liberal Democrats conveniently ignore..."). Human coder correctly identified this engagement and dismissal as degrading the counterargument (Level 1). Model incorrectly coded 0 (Ignored), failing to recognize the explicit engagement with the "fairness" argument.

B.22 Speech 19: d00128fdc5ad744da4f7abaa5f1de404

Mr. HATFIELD: Mr. President. I agree that Congress should increase the minimum wage standard. I have voted for reasonable minimum wage increases in the past and will certainly vote for the reasonable increase of the minimum wage today. As this Congress discusses welfare reform. it has been emphasized time and time again that those who can work should work. However. with the minimum wage today at \$4.25 an hour. a person laboring 8 hours a day. 5 days a week. 52 weeks a year would gross only \$8.840... If the minimum wage in January

1978 had kept pace with the Consumer Price Index... the current level would be \$6.40 in 1996. If we expect those on welfare to work. we can at least ensure that a minimum wage is a living wage and by voting for an increase in the minimum wage today we will have taken steps to assure those who are working are justly compensated for their work... A minimum wage provides vital protection for those workers who are not union members or who have few if any skills and little bargaining power. With bipartisan support. Congress should raise the minimum wage to \$5.15 per hour and I support that increase.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 2 | 2 | ✓Match |
| Justification-content | 1 | 2 | ~ Within one |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 2 | ~ Within one |
| Respect-counterarg | 0 | 1 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓Match |

B.22.1 Analysis:

Discrepancies on justification content, respect for demand, and counterargument respect.

Justification-Content : Human 1 (Self/group interest), model 2 (Common good - aggregate). Speaker argues the increase is needed to ensure a "living wage," provide "vital protection" for low-skill/non-union workers, ensure workers are "justly compensated," and align with the expectation that people should work. Model correctly identifies this focus on fairness, worker protection, and societal expectations as relating to common good/aggregate interests (Level 2). Human coding of 1 (Self/group interest) seems incorrect given the broad framing.

Respect-Demand : Human 1 (Neutral), model 2 (Explicitly respectful). Speaker states "I agree that Congress should increase...", calls the proposed increase "reasonable," and concludes "I support that increase." Model correctly identifies these explicit statements of agreement and support as explicitly respectful towards the demand (Level 2). Human code 1 (Neutral) appears incorrect here.

Respect-Counterarg : Human 0 (Ignored), model 1 (Degraded). The speech argues for the minimum wage increase but does not engage with any arguments against it. Human 0 (Ignored) is correct. Model 1 (Degraded) is incorrect.

B.23 Speech 20: 5bf3ed0e186078c7f98a7a9ddc717d8d

"Mrs. CLAYTON: They would also gain 300.000. 300.000 will gain."

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|------------|
| Participation | 1 | 1 | ✓ Match |
| Justification-level | 0 | 0 | ✓ Match |
| Justification-content | 1 | 1 | ✓ Match |
| Respect-group | 1 | 1 | ✓ Match |
| Respect-demand | 1 | 1 | ✓ Match |
| Respect-counterarg | 0 | 2 | × Disagree |
| Constructive Politics | 0 | 0 | ✓ Match |

B.23.1 Analysis:

One discrepancy on counterargument engagement in an extremely brief statement.

Respect-Counterarg : Human 0 (Ignored), model 2 (Acknowledged). This very short, factual statement ("300,000 will gain") offers no engagement whatsoever with counterarguments. Human code 0 (Ignored) is correct. Model code 2 (Acknowledged) is incorrect and likely a hallucination or error due to the brevity and lack of context.

B.24 Speech 21: bca7ea8c3a3783e41a4d7b55773a7f85

Mr. BIDEN: Mr. President. I listened with great interest to my friend on the committee. Senator GRASSLKY. who is one of the most productive members... with whom I vote a lot... I respectfully suggest. though. that the distinctions between the Biden provision on amending habeas corpus... and the Presidents are significant. but neither of which are going to make it fundamentally safer... If we eliminate habeas corpus... all of those people are already in Jail... The distinction between the Biden approach... and the Bush approach... the Biden approach on the death penalty... Racial Justice is sound... everyone knows the death penalty has been applied in ways that seem not to be fair... Unlike many people who support the racial justice provision. I am one who supports the death penalty... I am prepared to support the death penalty without the racial justice provision... Now. Mr. President. we all know what the vote is going to be here... I know where the votes are. There are not votes to sustain any position on racial justice. So let us have the vote and let us move on... My very distinguished colleague from Texas. Senator GRamm. says the President is entitled to a vote on his bill. Here. hear... let us give the President a vote on his bill. up or down... The Presidents bill consists of three things basically: the death penalty. which is in mine. habeas corpus. which is in mine... and the exclusionary rule. which is in mine... But that is all his bill is... my distinguished colleague from South Carolina... Senator THURMOND. has been asking me for 3 days. please give me a vote up or down... on the Presidents bill... So I have spent the last week running around the Democratic side saying... please do not [amend]. Let us make this clear... Let us find out how many votes the President has... But now I find my friend from South Carolina... has now taken the Biden bill... and attached it

to the Presidents bill... What is it. BushBiden? What is it called now?... Let us vote on racial justice so we can stop talking about it... Then let us go... to the Presidents bill... we will not lay one glove on it... we will vote... I hope that will be considered... Let me just note once again... it is not the Congress that has not been tough. it is the President that has not been tough... [Gives examples of unenforced laws/penalties] ... They are telling us that we are not tough on crime. We have given them 230 tough laws. Apply them... I have heard the President say. 100 days. pass my bill... The chairman of the Judiciary Committee... says please. Republicans. give us a vote on the Presidents bill... Please. Republicans. do not thwart the President. Give the Republican President a vote...

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓ Match |
| Justification-level | 3 | 2 | ~ Within one |
| Justification-content | 1 | 1 | ✓ Match |
| Respect-group | 1 | 1 | ✓ Match |
| Respect-demand | 1 | 1 | ✓ Match |
| Respect-counterarg | 3 | 2 | ~ Within one |
| Constructive Politics | 0 | 2 | × Disagree |

B.24.1 Analysis:

Discrepancies regarding justification level, counterargument respect, and constructive politics.

Justification-Level : Human 3 (Sophisticated), model 2 (Qualified). Speaker argues that habeas changes won't significantly improve safety, critiques the President's lack of enforcement using examples, and details complex legislative maneuvering around different bills and votes. Human coder likely saw these multiple distinct lines of argument (substantive and procedural) as sophisticated (Level 3). Model might have focused primarily on the procedural/legislative strategy aspects, perhaps seeing fewer distinct substantive justifications, thus coding qualified (Level 2).

Respect-Counterarg : Human 3 (Valued), model 2 (Acknowledged). Speaker uses respectful terms for colleagues ("my friend," "productive member," "distinguished colleague") while engaging with their positions (e.g., Gramm's call for a vote on the President's bill, Thurmond's request). Human coder weighed these parliamentary courtesies and the engagement with opponents' procedural requests as valuing (Level 3). Model likely focused more on the substantive criticisms (e.g., of the President's lack of enforcement, the limitations of the President's bill) and the description of tactical maneuvering, leading to a code of acknowledgment (Level 2) without net valuation.

Constructive Politics : Human 0 (Positional), model 2 (Mediating). Speaker describes a complex legislative deadlock and proposes specific procedural paths forward (vote on racial

justice first, then vote on an unamended President’s bill). Model likely interpreted these proposals for specific voting sequences as attempts to break the deadlock and find a way forward, classifying them as mediating proposals (Level 2). Human coder correctly identified this as strategic maneuvering within positional politics (trying to force votes or expose lack of support), not a compromise on the substance of the bills, hence positional (Level 0).

B.25 Speech 22: 7fd8bcfa874da662f816c87292dc1ea6

Mr. METZENBAUM: Mr. President. if you went out today to buy a lottery ticket. the odds are something like one million to one against you. Those odds are no greater than the odds you could have received 6 months ago that the distinguished minority leader and I would be joining together to offer the Brady amendment on the floor of the Senate. I am very pleased that it has occurred and that we have been able to work this matter out. Mr. President. I rise in support of this Brady bill amendment to the crime bill. It was over 4 years ago that I first introduced a bill to mandate a waiting period. In that time over 80.000 people have been killed with handguns. I do not rise to say that had the Brady bill been in effect since that time. all 80.000 people would not have been killed. Of course not. I think that a substantially lesser number would have been killed. This past January. I again introduced this bill. S. 257. along with Senators KAssEBAUM. ChAFEi. KENNEDY.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓ Match |
| Justification-level | 3 | 1 | × Disagree |
| Justification-content | 1 | 1 | ✓ Match |
| Respect-group | 1 | 1 | ✓ Match |
| Respect-demand | 1 | 1 | ✓ Match |
| Respect-counterarg | 0 | 1 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓ Match |

B.25.1 Analysis:

Significant discrepancy on justification level and disagreement on counterargument respect.

Justification-Level : Human 3 (Sophisticated), model 1 (Inferior). Speaker supports the Brady bill using a statistic (80,000 killed) and a qualified claim about its potential impact ("substantially lesser number would have been killed"). Human coder apparently interpreted this combination, perhaps along with the context of bipartisan effort mentioned earlier, as a sophisticated justification (Level 3). This seems overly generous for the provided text. Model saw only a statistic paired with a qualified, speculative claim, rating it as inferior justification (Level 1). The model’s coding appears more aligned with the text snippet than the human’s.

Respect-Counterarg : Human 0 (Ignored), model 1 (Degraded). Speaker argues in favor of the Brady bill but does not mention or engage with arguments against it. Human 0 (Ignored) is correct. Model 1 (Degraded) is incorrect.

B.26 Speech 23: 715ecdc2e373327567f6fadecb467cb7

Mr. GEKAS: Mr. Chairman. the bill that the Judiciary Committee fashioned... bears no resemblance to the work that has been done by the last several Congresses. For example. the Senate... passed a comprehensive crime bill which included substantially those provisions which the President offered with respect to habeas corpus. the death penalty... But the bill that was fashioned by the Democratic majority... leans heavily toward the protection of the individual who is convicted of murder... First of all. the original bill. as fashioned by the President... is totally different than the substantive notions upon which the death penalty can be set. For instance. the bill that is before us says that the death penalty by and large must be relegated to only those killings which come about with an intent to kill. but we say. of course. an intent to kill should give the jury the right... But we also say that when an actor... acts with reckless disregard or with reckless indifference to life... like a driveby killing... we believe... that individual should be just as subject to the possibility of the death penalty... That is an outstanding difference... Why is that substantial? Because last years bill rested on that very same point... We then fought the entire battle last year on that point. We prevailed... Here we are again. but here is a significant difference that proves our point by the exception that the majority put in their bill... they put in a provision that says a driveby killing... What does that mean? It means they drew it so narrowly... that if the individual... steps out of that car... he would not be subject to the death penalty uider the provisions that the majority Democrats... imposed... So the answer is to impose our standard... That is a substantial difference... If for any reason you feel that the majority bill is adequate. you should reject it on that basis alone... I ask the Members to support. when the time comes. the Gekas amendment...

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 3 | 2 | ~ Within one |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 1 | ✓Match |
| Respect-counterarg | 0 | 1 | ~ Within one |
| Constructive Politics | 1 | 0 | ~ Within one |

B.26.1 Analysis:

Discrepancies on justification level, counterargument respect, and constructive politics.

Justification-Level : Human 3 (Sophisticated), model 2 (Qualified). Speaker provides a detailed critique comparing the committee bill’s death penalty standard (intent to kill) with his preferred standard ("reckless disregard"), using legislative history ("last year’s bill," "We prevailed") and a specific critique of the drive-by killing exception ("drew it so narrowly," hypothetical of stepping out of car). Human coder likely saw this detailed, multi-faceted critique and proposal as sophisticated (Level 3). Model might have interpreted this as primarily elaborating on one core point (the standard for the death penalty), thus coding it as qualified (Level 2).

Respect-Counterarg : Human 0 (Ignored), model 1 (Degraded). Speaker critiques the committee bill ("leans heavily toward protection of the individual," "drew it so narrowly") and contrasts it with his preferred standard, but does not engage with the reasons or arguments proponents of the committee bill might offer for their approach. Human 0 (Ignored) is correct. Model 1 (Degraded) is incorrect.

Constructive Politics : Human 1 (Alternative), model 0 (Positional). Speaker advocates for replacing the committee’s standard with "our standard" via the Gekas amendment. Human coder interpreted this push for a distinct alternative standard (via amendment) as an alternative proposal (Level 1). Model likely saw advocating for one’s own amendment over the main bill as standard positional politics (Level 0). The model’s interpretation is reasonable, as advocating for an amendment is typical positional behavior unless the amendment represents a significant bridging or novel approach not previously on the agenda.

B.27 Speech 24: f4380f373997d3cfa8c408faf9286321

Mr. KENNEDY: I will Just take 2 minutes. Mr. President. on the one hand. the Senator from Utah says there are only going to be a few hundred thousand that are going to be affected. The Senator cannot have the argument both ways. He cannot say there is only a few hundred thousand going to be affected and then predict these dramatic and dire economic Impacts on the economy. The argument does not hold. It just does not hold.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 2 | 1 | ~ Within one |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 1 | ✓Match |
| Respect-counterarg | 1 | 2 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓Match |

B.27.1 Analysis:

Disagreements on justification level and counterargument respect.

Justification-Level : Human 2 (Qualified), model 1 (Inferior). Speaker points out an apparent internal contradiction in the opponent's argument ("cannot have the argument both ways": low impact number vs. dire economic prediction). Human coder saw identifying and explaining this logical inconsistency as providing a qualified reason (Level 2) to dismiss the opponent's point. Model might have interpreted this as merely stating the contradiction ("argument does not hold") without fully elaborating why the combination is invalid, thus coding it as inferior (Level 1).

Respect-Counterarg : Human 1 (Degraded), model 2 (Acknowledged). Speaker directly addresses the opponent's specific argument ("the Senator from Utah says...") and then dismisses it forcefully ("cannot have the argument both ways," "argument does not hold. It just does not hold."). Human coder correctly identified this direct refutation and dismissal as degrading the counterargument (Level 1). Model coded 2 (Acknowledged), likely focusing on the direct attribution ("the Senator... says") as acknowledgment but failing to register the subsequent dismissal as degradation.

B.28 Speech 25: d8573836762a83df5e1a2b568fe5fb8a

Mr. SMITH of Texas: Mr. Chairman. I thank my friend and colleague for yielding me this time. Mr. Chairman. last week President Bush said that our civil rights laws should not encourage litigation at the expense of conciliation. mediation. or settlement. Yet if we dont adopt the Sensenbrenner amendment. that is exactly what could happen to the ADA. First let me stress that the Sensenbrenner amendment guarantees to victims of discrimination the same rights and remedies that are available to other protected classes under title VII... It is simply designed to state that Congress believes that title VII remedies are appropriate... Many people do not realize that the actual award given in a lawsuit is just the tip of the iceberg in terms of the total litigation costs. The more the incentive to litigateas would surely be the case if the ADA is expanded to include punitive and compensatory damages the higher the litigation costs... Under title VII with its current remedies. only 6 percent of the charged filed with the EEOC end up as lawsuits... Of those cases that go to trial... the attorneys fees average more than \$96.000 per case. A recent study... estimates that title VII litigation will triple if it includes punitive and compnensatory damages. If ADA is included... even more lawsuits are sure to result. The ADA with the addition of the Sensenbrenner amendment. will assure that the law will open the door to the workplace. rather than the door to the courthouse. Mr. Chairman. a vote for the Sensenbrenner amendment Is a vote for continuing an effective policy... I urge its support:

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 3 | 2 | ~ Within one |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 1 | ✓Match |
| Respect-counterarg | 0 | 1 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓Match |

B.28.1 Analysis:

Disagreements on justification level and counterargument respect.

Justification-Level : Human 3 (Sophisticated), model 2 (Qualified). Speaker argues for the Sensenbrenner amendment by quoting President Bush, drawing an analogy to Title VII remedies, providing statistics on current litigation rates and costs, citing a study projecting increased litigation, and framing the amendment as promoting workplace access over court access. Human coder likely saw these multiple, linked lines of reasoning (authority, analogy, statistics, projection, framing) as sophisticated (Level 3). Model might have viewed this as primarily one justification (avoiding increased litigation costs) supported by various pieces of evidence, thus coding it as qualified (Level 2).

Respect-Counterarg : Human 0 (Ignored), model 1 (Degraded). The speaker argues for the Sensenbrenner amendment (which limits remedies) but does not engage with the arguments against it or for the alternative approach of allowing expanded damages under the ADA. Human 0 (Ignored) is correct. Model 1 (Degraded) is incorrect.

B.29 Speech 26: f33f20115382484a57af2f00388079b7

Mr. MURKOWSKI: Mr. President. I rise today to voice my total opposition to the gun control provisions contained in S. 1241. The so-called assault weapon ban and the Brady bill will do nothing to reduce crime and will only penalize legitimate gun owners. Title VII... would ban the transfer, import, or possession of any assault weapon... Proponents of this provision would have us believe that it is aimed at controlling crime... They would have us believe that these weapons are the weapon of choice for criminals... They would have us believe that by banning the possession... we would somehow reduce crime. They are wrong. The people affected by this gun control measure would be the law-abiding citizens... the collectors, the target shooters, the hunters... they deserve our respect. They deserve measures that will truly reduce crime. They do not deserve to be scapegoats... why will a ban on assault weapons not reduce violent crime...? It is because, by definition, criminals commit crimes... Criminals do not obey the law... FBI 1988 statistics show that only 4 percent of homicides were committed with rifles... S. 1241 would call for a study... if this study shows that this ban

is ineffective... those supporting the ban today will not then be calling for Its repeal. No. their purpose is not to reduce crime. it is to reduce gun ownership... It is an effort to penalize... This gun control language is only the first step toward eliminating our second amendment rights... This attack on honest gun owners is a twopronged attack. S. 1241 also includes the famous Brady bill... contains two basic provisionsone to mandate a 7day waiting period... and one to mandate a background check... As with the assault weapon ban. there is no evidence that a waiting period will help reduce crime... evidence overwhelmingly points to waiting periods having no effect... Several States currently have waiting periods... New York. New Jersey... yet they have experienced no corresponding decrease in crime... Criminals are not going to wait 7 days... Most guns used in crimes are obtained illegally... Again. the only people affected... will be lawabiding citizens... How about this laws effect on crimes of passion... a crime of impulse will be committed with whatever weapon is closest at handnot after a trip to a gun store... I am pleased to support the amendment of my colleague... Let me discuss... the effect of this legislation on the State of Alaska... high percentage... live in rural areas... need to fly... hundreds of miles to the nearest gun store... forced to sit around in a hotel for 7 days. That is absolutely ridiculous... Why do Alaskans need handguns? One reason is for hunting... The major reason is for protection... The second aspect of the Brady bill... the mandatory check... This provision makes more sense... However. the Brady bill ties this provision with the 7day waiting period instead of mandating an Instant check... Seven days is not enough time to do a thorough paperbased background check... In closing. Mr. President. I would like to simply say that the gun control measures included in this act are a smokescreen... If we are serious about reducing crime I suggest we leave off discussions of these ineffectual measures and concentrate on criminals. -not honest gun owners.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 4 | 3 | ~ Within one |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 0 | 0 | ✓Match |
| Respect-counterarg | 2 | 1 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓Match |

B.29.1 Analysis:

Discrepancies on justification level and counterargument respect.

Justification-Level : Human 4 (In-depth - Original Scale), model 3 (Sophisticated-broad). This is a very long speech presenting numerous arguments against both the assault weapon ban and the Brady bill's waiting period. Arguments include: criminals don't obey

laws, impact on law-abiding citizens (collectors, hunters, Alaskans), FBI statistics, ineffectiveness of waiting periods (state examples, illegal acquisition, crimes of passion), preference for instant checks, motive attribution (reduce gun ownership not crime), framing as "scapegoats" and "smokescreen." Human coder saw this extensive, multi-pronged attack as in-depth (Level 4). Model identified multiple linked arguments, fitting sophisticated (Level 3), but perhaps didn't reach the threshold for in-depth integration or complexity. A close call.

Respect-Counterarg : Human 2 (Acknowledged), model 1 (Degraded). Speaker explicitly states the arguments of proponents ("Proponents... would have us believe...") and then strongly refutes them ("They are wrong"). He also attributes negative motives ("their purpose is not to reduce crime, it is to reduce gun ownership," "effort to penalize," "attack," "smokescreen"). Human coder focused on the explicit statement of the opponent's claims as acknowledgment (Level 2). Model likely weighed the strong refutation ("wrong"), the negative framing ("scapegoats," "smokescreen"), and the attribution of ulterior motives more heavily, resulting in a degraded classification (Level 1). Given the dismissive language and motive attribution, the model's interpretation seems more accurate here.

B.30 Speech 27: a5a2a0dc010107434047d25c6fd01e2e

Mr. PAYNE of Virginia: Mr. Chairman. I rise in support of Mr. OLINS amendment. I strongly support the purposes of the ADA bill. yet despite all the effort expended on its development. there still remain portions which provide differences in interpretation. such as the section concerned with accommodation to facilitate the employment of a disabled job applicant. Once this bill becomes public law. it will be necessary for employees across the country to make millions of individual decisions as to how it might be implemented. "Undue hardship" and "reasonable accommodation" will be the only guidelines available to them. with no assurance or certainty that whatever they propose by way of accommodation will not lead to litigation. and subsequent lengthy judicial resolution. I believe the Olin amendment. provides defined. reasonable standardsreasonable to those who are disabled. while at the same time reasonable to the employers. I urge our colleagues to support this amendment. and thereby facilitate the movement into the workforce of the disabled who seek employment.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 2 | 2 | ✓Match |
| Justification-content | 3 | 1 | × Disagree |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 1 | ✓Match |
| Respect-counterarg | 0 | 1 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓Match |

B.30.1 Analysis:

Disagreement on justification content and counterargument respect.

Justification-Content : Human 3 (Universal principles), model 1 (Self/group interest). Speaker argues the Olin amendment provides needed clarity ("defined, reasonable standards") to avoid uncertainty and litigation associated with terms like "undue hardship." He frames these standards as "reasonable to those who are disabled" and "reasonable to the employers," facilitating workforce entry. Human coder focused on the appeal to reasonableness, clarity, and fairness for both groups, aligning with universal principles (Level 3). Model likely keyed specifically on the explicit mentions of the groups "disabled" and "employers," interpreting the argument as balancing their respective interests, thus coding self/group interest (Level 1).

Respect-Counterarg : Human 0 (Ignored), model 1 (Degraded). Speaker argues for the Olin amendment by highlighting the perceived problems (ambiguity, litigation risk) with the existing bill's terms ("Undue hardship," "reasonable accommodation"). However, the speaker does not engage with any arguments against the Olin amendment itself. Human 0 (Ignored) is correct. Model 1 (Degraded) is incorrect.

B.31 Speech 28: f38f0d305f72c52ea5098a937002f6d9

Mr. McMILLEN of Maryland: I rise in favor of the Volkmer amendment striking provisions banning the ownership of certain firearms. I have heard from a number of my constituents on this issue. and agree with them that in our efforts to control crime. we need not infringe upon the legitimate rights of individuals. I am concerned about the proliferation of urban crime. and the attendant violence. We need to control the illegal use of these guns. and keep them out of the hands of criminals. Unfortunately. the ban proposed in the legislation is imprecise. it denies legitimate uses of certain guns. and would negatively impact domestic manufacturers. More important. the presupposition of the legislation is that everyone who purchases certain rifles is either deranged or a potential criminal. This clearly is not the case. If we really want to address the issue of criminal use of semiautomatic weapons and to limit violence. we should not try to simply ban certain categories of rifles. but. rather. look into other alternatives such as registration or waiting periods. Previously. I have supported a waiting period for handguns. and see potential in extending this to other areas. The effectiveness of this approach has been demonstrated by various States. including my home State of Maryland. In summary. though the aims of the legislation are admirable. there are better ways of accomplishing the same goals.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓ Match |
| Justification-level | 2 | 2 | ✓ Match |
| Justification-content | 0 | 2 | × Disagree |
| Respect-group | 1 | 1 | ✓ Match |
| Respect-demand | 2 | 1 | ~ Within one |
| Respect-counterarg | 0 | 2 | × Disagree |
| Constructive Politics | 1 | 1 | ✓ Match |

B.31.1 Analysis:

Disagreements on justification content, respect for demand, and counterargument respect.

Justification-Content : Human 0 (None), model 2 (Common good - aggregate). Speaker argues against the proposed ban because it's "imprecise," denies "legitimate uses," harms manufacturers, relies on a false presupposition about buyers, and isn't the best way to "control crime" or "limit violence." He suggests alternatives (registration, waiting periods) aimed at these goals. Model likely interpreted the focus on controlling crime/violence and finding effective policy solutions as an appeal to common good/aggregate interests (Level 2). Human coding of 0 (None) is puzzling, as the speaker clearly provides reasons for his position linked to broader concerns (crime, rights, effectiveness).

Respect-Demand : Human 2 (Explicitly respectful), model 1 (Neutral). Speaker explicitly states that the "aims of the legislation [he opposes] are admirable." Human coder correctly identified this positive characterization of the opponent's intent as explicit respect (Level 2). Model coded neutral (Level 1), failing to capture this specific instance of expressed respect for the underlying goals of the opposed demand.

Respect-Counterarg : Human 0 (Ignored), model 2 (Acknowledged). Speaker argues against the ban proposed in the legislation but does not engage with the specific arguments or reasons put forth by proponents of the ban. He critiques the ban itself and its presuppositions. Human 0 (Ignored) is correct. Model 2 (Acknowledged) is incorrect.

B.32 Speech 29: 05ab74aa2d5521539590b352424ca0a1

Mr. MORAN: Mr. Chairman. almost all of us could relate to Mr. and Mrs. Collins and the pride and the love that they felt for their daughter. She was beautiful. bright. accomplished. athletic. full of spunk and fun. Until July 12. 1985. when she was jogging. and a man grabbed her from behind... smashed her face... sexually tortured her. put a branch through the length of her body repeatedly until she died... There was no question of guilt here... he confessed... claimed multiple personality disorders... unanimous conclusion was that he did not suffer from any isms but he was a very cruel person. I am not going to go any further into this persons background because under current law and under

this House bill it is simply too easy to find reasons for further Federal appeals. This was 6 years ago. No question of guilt. I cannot describe in detail what happened to the Collins daughter. but her father has had to do this for 6 years. will have to do that for years more. What punishment. what revictimization? I have heard these arguments. and I want to believe and agree with my side of the aisle. I cannot in this case. The point is that he was arrested over 6 years ago. It took 20 more months to get to trial. He was convicted 4Y2 years ago and since then has continually been able to appeal. It will be years of more hell for Mr. and Mrs. Collins under current law and under this bill because this bill does not set adepuate time limits... The Hyde amendment on the other hand sets a statute of limitations. I have to make the point that it is time we brought the punishment of innocent survivors to closure. It is not fair what we are doing.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 2 | 2 | ✓Match |
| Justification-content | 1 | 1 | ✓Match |
| Respect-group | 1 | 2 | ~ Within one |
| Respect-demand | 1 | 1 | ✓Match |
| Respect-counterarg | 2 | 1 | ~ Within one |
| Constructive Politics | 0 | 0 | ✓Match |

B.32.1 Analysis:

Discrepancies regarding respect for group and counterarguments.

Respect-Group : Human 1 (Neutral), model 2 (Explicitly respectful). Model likely coded the highly empathetic and detailed description of the Collins family’s ordeal ("pride and the love," "hell for Mr. and Mrs. Collins," "punishment of innocent survivors") as expressing explicit respect and value for this group (victims’ families) (Level 2). Human coder perhaps focused more narrowly on groups directly involved in the legislative debate (proponents/opponents of amendments), coding neutral (Level 1) if no explicit respect was shown to those specific groups.

Respect-Counterarg : Human 2 (Acknowledged), model 1 (Degraded). Speaker mentions hearing arguments from "my side of the aisle" ("I have heard these arguments, and I want to believe and agree...") but ultimately rejects them in this specific case ("I cannot in this case"). Human coder interpreted mentioning these arguments as acknowledgment (Level 2). Model might have interpreted the strong emotional weight of the narrative and the subsequent rejection of his own side’s arguments as implicitly degrading those (unstated) counterarguments by prioritizing the victim’s story and the need for closure (Level 1).

B.33 Speech 30: 5fd8b62c9841898d460b90fe25f4d951

Mr. HOUGHTON: Mr. Speaker. in the great debate over the minimum wage... one fact is quite well camouflaged. Raising the minimum wage does not help the working poor the way we think It will. Minimum wage earners make up less than 4 percent of our national workforce... Of the 20 million Americans... in poverty. only about 335.000 households earn the minimum wage... So by beating the drums... on...minimum wagewe avoid the real problem... What Id like to do is to try to untangle ourselves... and look at another approachone championed by my colleague... TOM PETRI... This approach is called EITCorthe Earned Income Tax Credit. What this would do is allow disadvantaged families earning approximately the minimum wage level to receive a tax credit... [Details EITC structure and benefits] ... There are many approaches to this problem of povertyand help for those who need help. Personally. I think going the EITC route is a fNr way to look at the condition... Should the minimum wage be raisedyes. but not to the point of jeopardizing th.. job producers. If we are serious about helping the working poor there is another way... What EITC does is push aid to the less fortunate in a way that frankly provides support for those who choose to work rather than those who dont... EITC will provide those who wish to work... with an incentive to keep trying... The result would be a tax credit costing the Government far lessmaybe nothing at all... a reduction in the number of welfare recipients. combined with the taxes generated... would make the program selfsufficient... The Congress needs to consider this idea... This is why Ive decided to cosponsor the Family Living Wage Act... It Is Innovative. straightforward. uncomplicated and. best of all. It gets to the heart of the real problem.

| Dimension | Human Coding | Model Coding | Agreement |
|-----------------------|--------------|--------------|--------------|
| Participation | 1 | 1 | ✓Match |
| Justification-level | 3 | 3 | ✓Match |
| Justification-content | 1 | 2 | ~ Within one |
| Respect-group | 1 | 1 | ✓Match |
| Respect-demand | 1 | 1 | ✓Match |
| Respect-counterarg | 0 | 2 | × Disagree |
| Constructive Politics | 1 | 1 | ✓Match |

B.33.1 Analysis:

Disagreements on justification content and counterargument respect.

Justification-Content : Human 1 (Self/group interest), model 2 (Common good - aggregate). Speaker advocates for the Earned Income Tax Credit (EITC) as a superior approach to helping the "working poor" compared to minimum wage increases alone. He argues it targets aid better, helps "disadvantaged families," supports work incentives ("support for those who choose to work"), reduces welfare dependency, and potentially costs less. Model correctly identifies this focus on broad societal benefits (poverty reduction, work incentives,

welfare reduction, fiscal impact) as an appeal to common good/aggregate interests (Level 2). Human coding of 1 (Self/group interest) seems incorrect given the framing around helping the poor and improving societal outcomes.

Respect-Counterarg : Human 0 (Ignored), model 2 (Acknowledged). Speaker critiques the focus on minimum wage as potentially avoiding "the real problem" and being less effective than EITC. However, while acknowledging the minimum wage debate exists, he does not engage with the specific arguments for a significant minimum wage increase itself; instead, he pivots to promoting EITC as a better alternative. Human 0 (Ignored) is correct because the arguments for the minimum wage (beyond a modest increase) are not addressed. Model 2 (Acknowledged) is incorrect; critiquing the focus is not the same as acknowledging the substance of the opposing arguments.

B.34 Summary of Error Patterns (Model vs. Human Gold Standard)

Analysis of the 30 sample speeches reveals several recurring patterns of disagreement between the `gemini-2.0-flash` model annotations and the expert human coding:

1. Respect for Counterarguments (**Respect-Counterarg**) - Most Frequent Discrepancy: This dimension showed the most consistent and significant errors.

- **Failure to Detect Ignored Counterarguments:** The model frequently coded counterarguments as being 'Acknowledged' (2) or even 'Degraded' (1) when the human coder correctly identified them as 'Ignored' (0). The model seems to misinterpret general critique of an opposing *position* or *policy* as engagement with the specific *arguments* supporting that position.
- **Misclassification of Engagement Type:** When counterarguments *were* engaged, the model often misjudged the nature of that engagement:
 - It under-detected Degradation (coding 2 instead of 1), often classifying direct dismissals ("argument doesn't hold," "wrong") or strong refutations as mere Acknowledgment.
 - It struggled to identify Valuation (coding 1 or 2 instead of 3), especially when speakers combined standard parliamentary courtesy ("my distinguished colleague") with strong substantive disagreement or criticism. The model seemed to weigh the disagreement more heavily than the explicit respect shown.

2. Justification Content (**Justification-Content**) - Interpretation Differences: Disagreements were frequent here, often involving a full category mismatch (× Disagree).

- **Over-reliance on Group/Self-Interest Heuristics:** The model frequently defaulted to 'Self/group interest' (1) when human coders identified appeals based on 'Common good - aggregate' (2) or 'Universal principles' (3). This often occurred

when speakers mentioned specific groups (e.g., "small businesses," "disabled," "employers," "my state") or costs, even if the overarching argument concerned broader societal benefits like job creation, public safety, fairness, or economic justice.

- **Missed Appeals:** Occasionally, the model coded 'None' (0) when human coders perceived an implicit appeal to common good or principles.

3. **Justification Level (**Justification-Level**) - Detail and Structure:** Disagreements were common, though often 'within one' level ($\sim$ Within one).

- **Sensitivity to Complexity:** The model seemed less sensitive to the higher levels of justification complexity ('Sophisticated' 3 vs. 'In-depth' 4), often coding complex, multi-faceted arguments as Level 3 when humans saw Level 4.
- **Requirement for Explicit Structure:** The model sometimes required a clearer, more explicit premise-conclusion structure than human coders, leading it to code 'Inferior' (1) or 'None' (0) when humans saw 'Qualified' (2) justification based on anecdotes, statistics used implicitly, or quoted authority without speaker elaboration.

4. **Respect for Demand (**Respect-Demand**) - Valence of Direct Stance:** Disagreements often occurred around the neutral category.

- **Under-detection of Disrespect:** The model often coded 'Neutral' (1) when speakers used strong negative language directly against the policy proposal ("strong opposition," "harmful," "bad idea"), which humans coded as 'Disrespectful' (0).
- **Under-detection of Explicit Respect:** Similarly, the model sometimes coded 'Neutral' (1) despite explicit endorsements ("I support," "reasonable," "meritorious"), which humans coded as 'Explicitly Respectful' (2).

5. **Constructive Politics (**Constructive Politics**) - Procedural vs. Substantive:** Less frequent but notable disagreements occurred.

- **Over-Interpretation of Alternatives:** The model occasionally classified procedural suggestions (e.g., specific voting orders) or standard amendment advocacy as 'Alternative' (1) or 'Mediating' (2) proposals, whereas humans saw these as tactical moves within 'Positional' (0) politics.

6. **Respect for Group (**Respect-Group**) - Scope and Empathy:** Occasional 'within one' disagreements.

- **Missed Explicit Respect:** The model sometimes missed explicit statements of value or empathy towards affected groups (e.g., victims' families, low-wage workers), coding 'Neutral' (1) when humans coded 'Explicitly Respectful' (2). This may relate to focusing on political actors vs. broader societal groups.

Overall Assessment: The model performed well on the basic `Participation` dimension. However, it consistently struggled with dimensions requiring careful interpretation of rhetoric, pragmatic understanding (like parliamentary courtesy), distinguishing critique of policy from engagement with arguments, and abstracting broader principles from specific examples or keywords. The `Respect-Counterarg` dimension stands out as needing significant improvement. It's also worth noting that a small number of discrepancies might be attributable to potential human coder inconsistencies or ambiguity in the source text/coding scheme, especially for very short snippets.